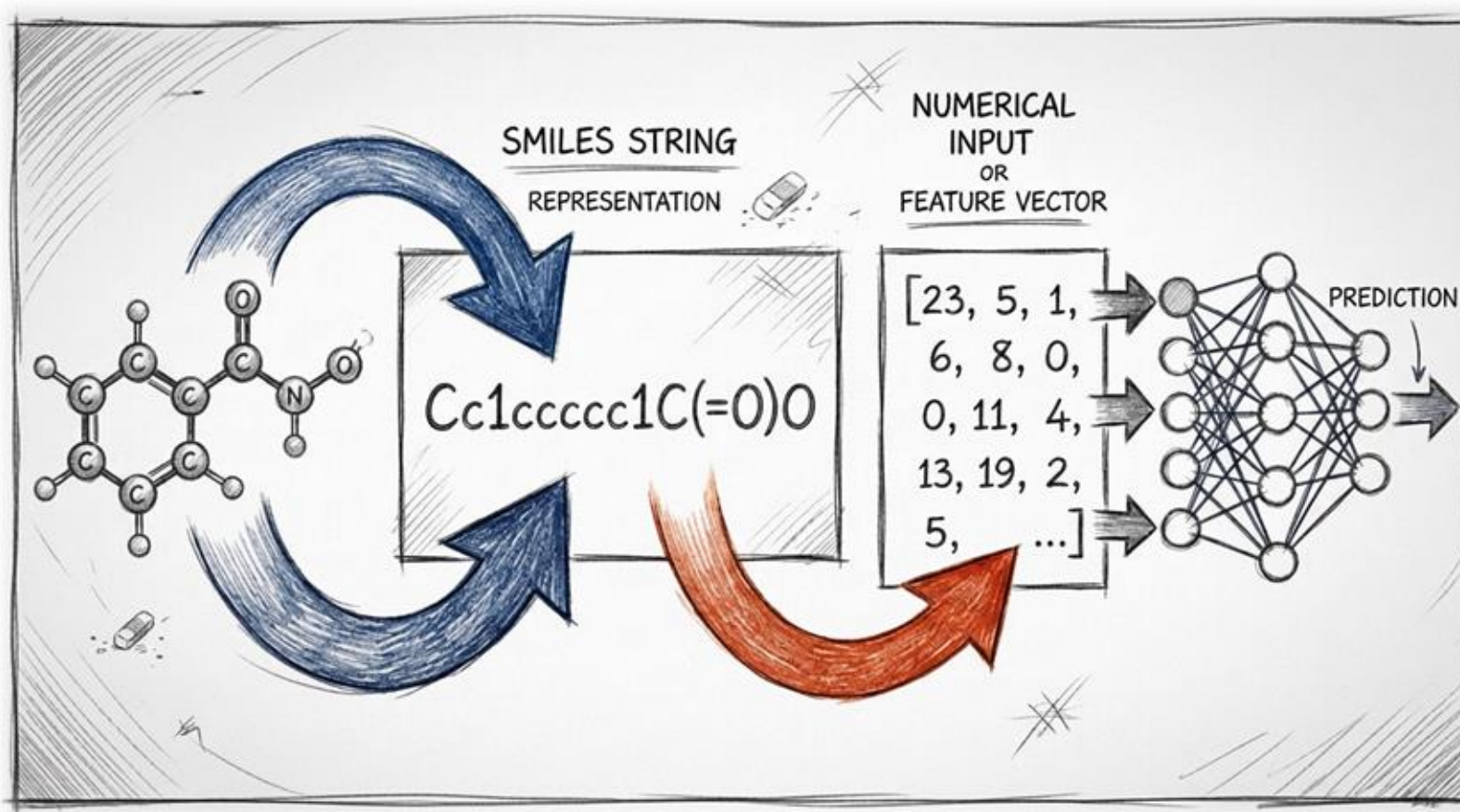


Chemical Representations for Machine Learning



Plan

- **Evolution of “data-driven” chemistry**
- Overview of existing representations
- Importance of the choice of the representation
- Data preprocessing: practical considerations
- Data visualization

Evolution of Data-Driven Chemistry

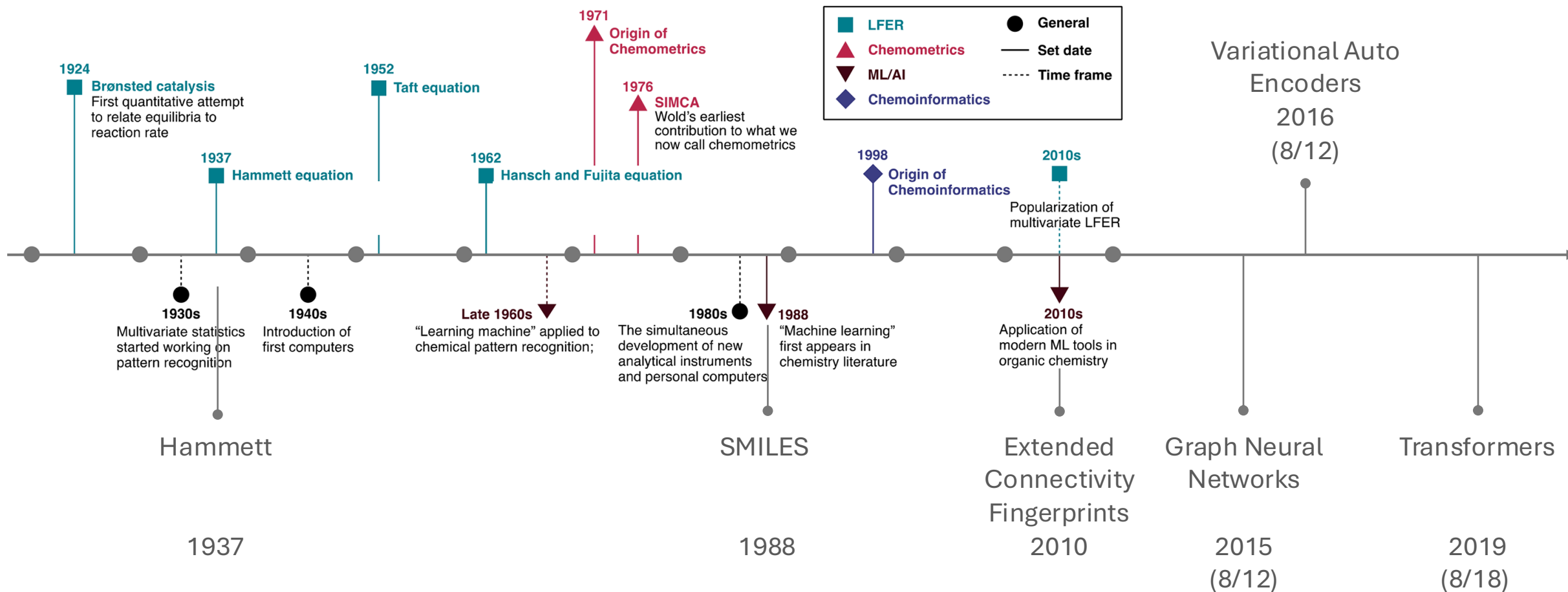
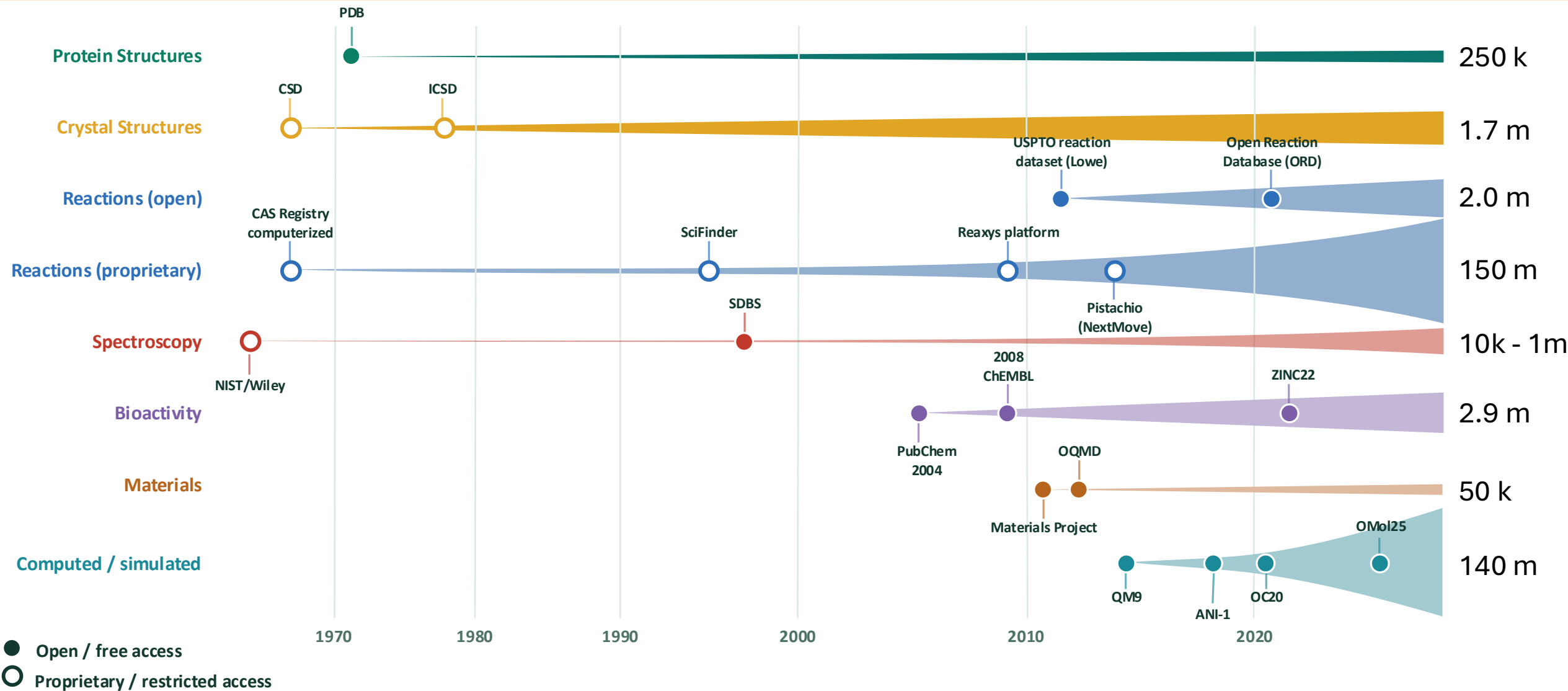
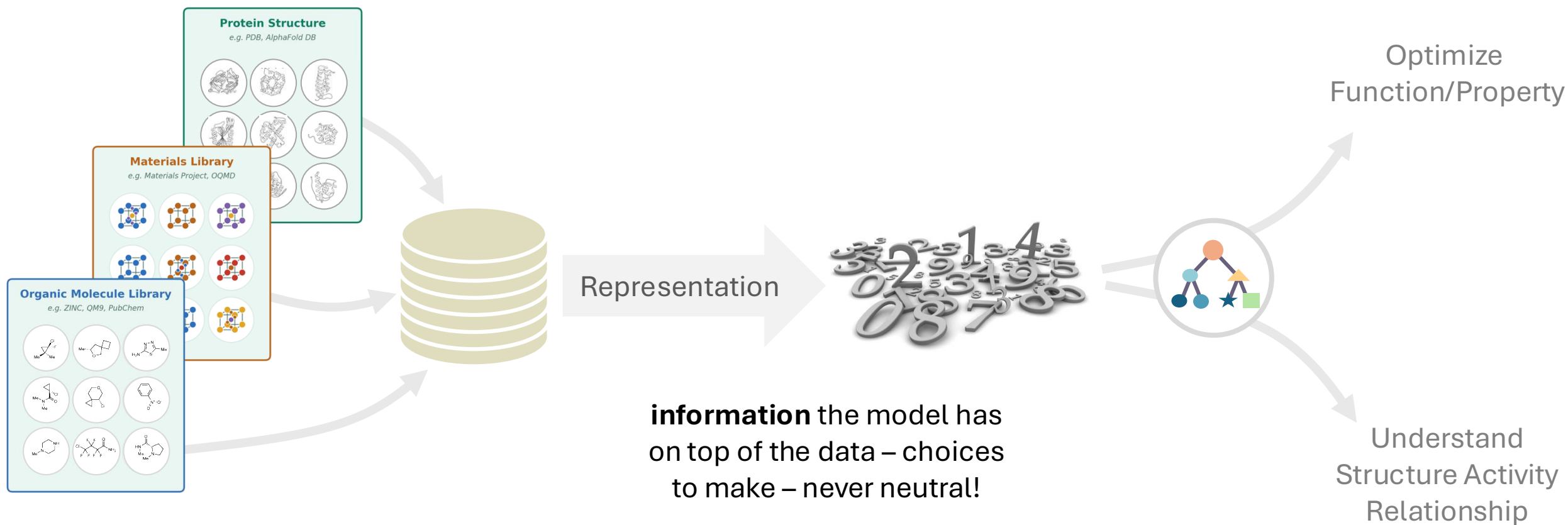


Figure adapted from Williams, W. L.; Zeng, L.; Gensch, T.; Sigman, M. S.; Doyle, A. G.; Anslyn, E. V. *ACS Cent. Sci.* **2021**, 7 (10), 1622–1637

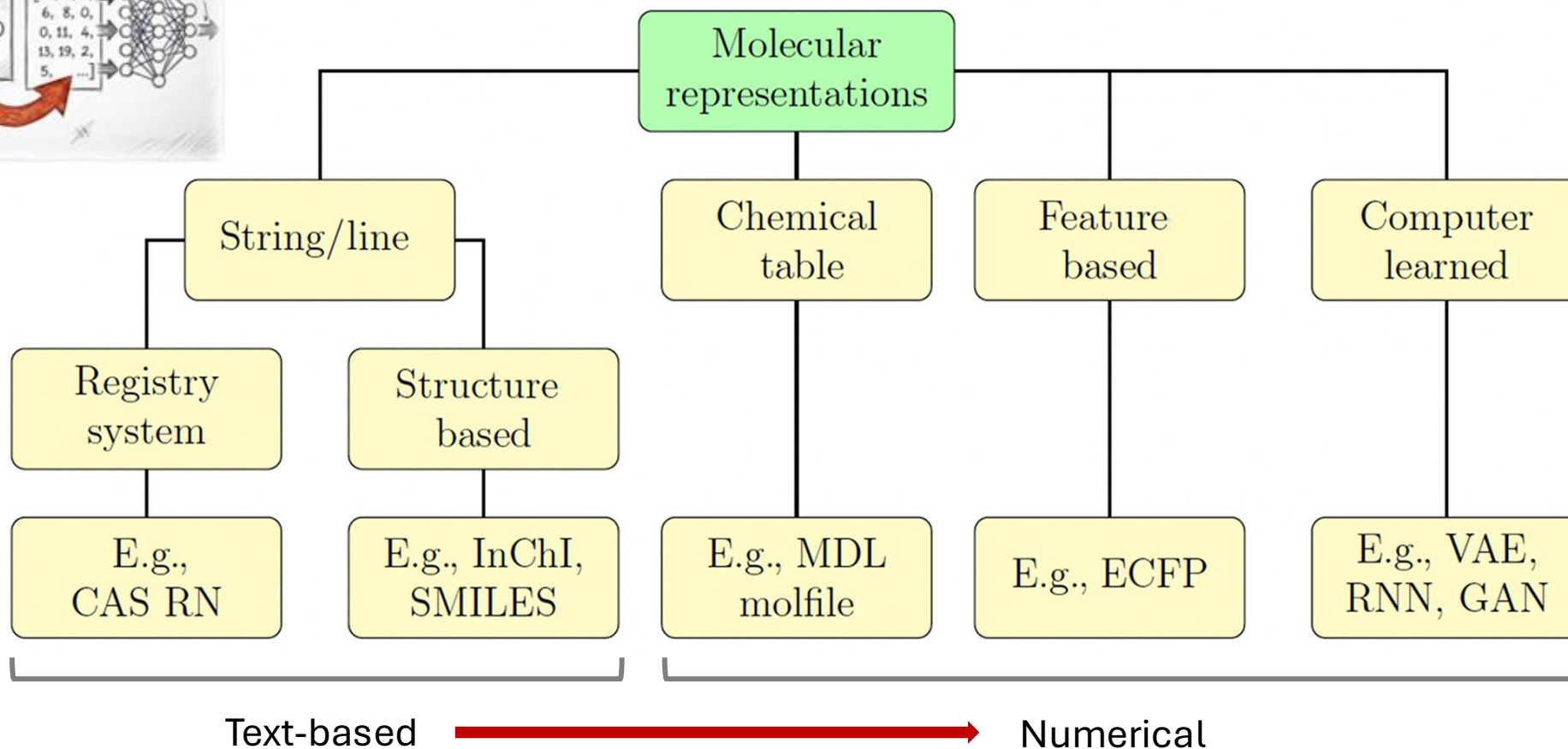
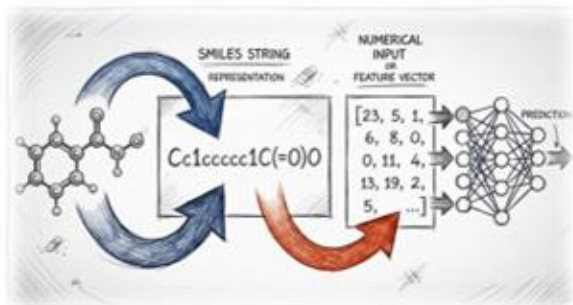
History of open-source datasets availability



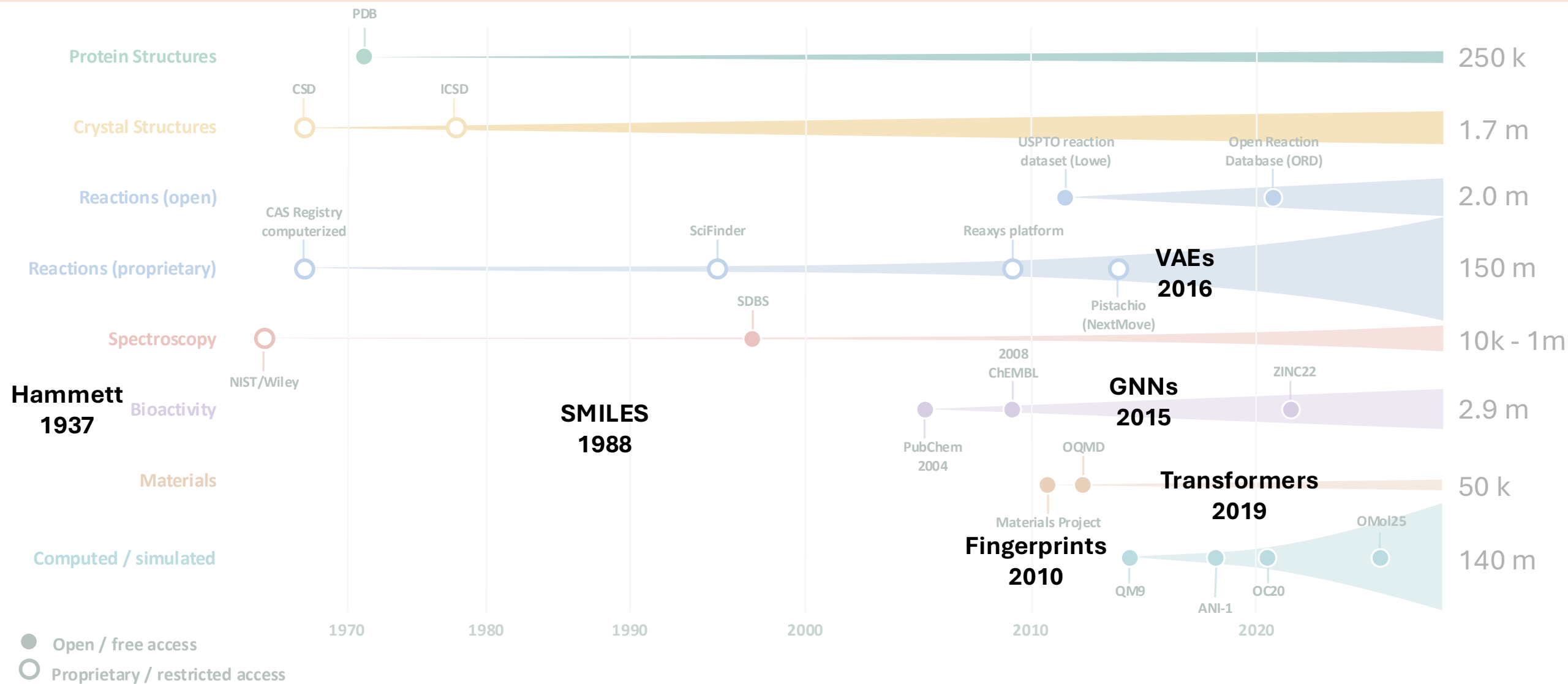
The need for molecular representations



Type of molecular representations



History of open-source datasets availability



The Evolution of Data-driven Chemistry

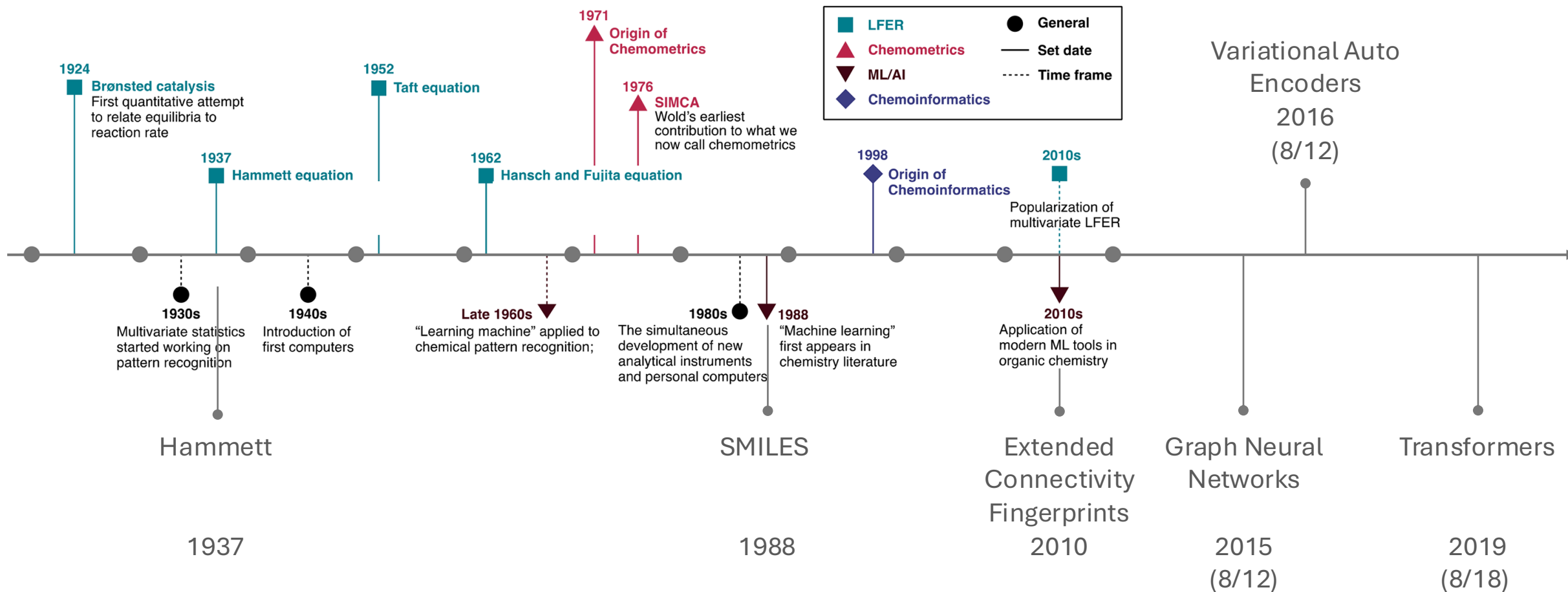
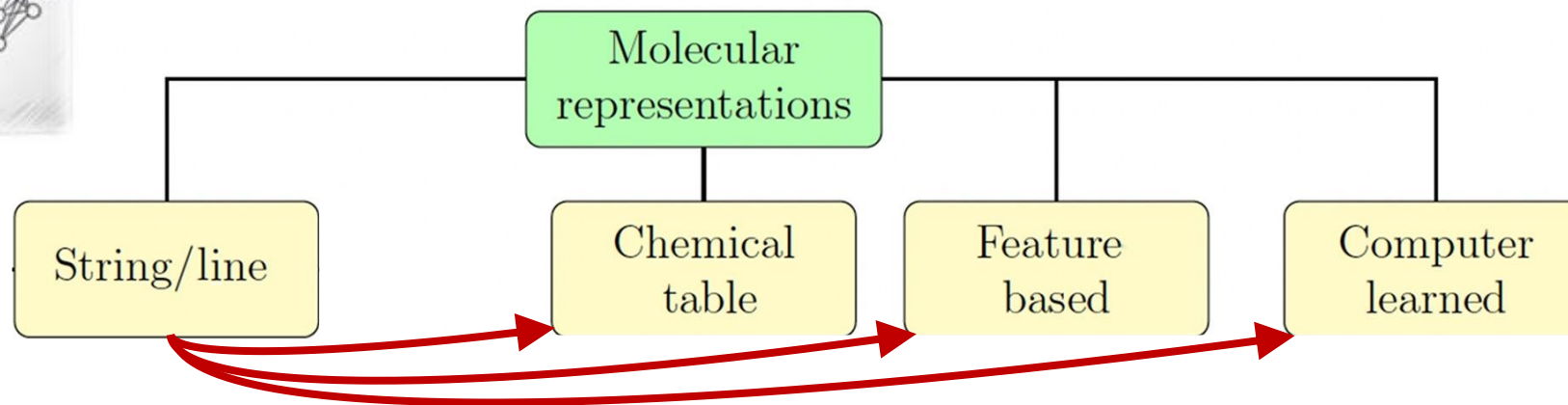
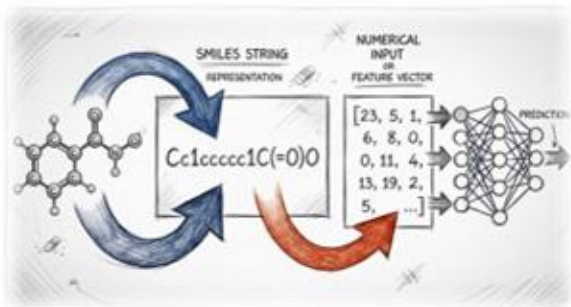


Figure adapted from Williams, W. L.; Zeng, L.; Gensch, T.; Sigman, M. S.; Doyle, A. G.; Anslyn, E. V. *ACS Cent. Sci.* **2021**, 7 (10), 1622–1637

Plan

- Evolution of “data-driven” chemistry
- **Overview of existing representations**
- Importance of the choice of the representation
- Data preprocessing: practical considerations
- Data visualization

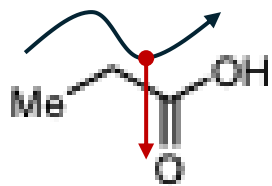
String based representations are usually the starting point



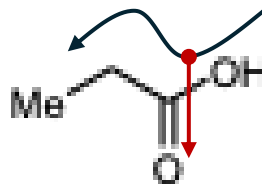
Small size
Easy to use
Potential missing Information

String line representations for small molecules: SMILES

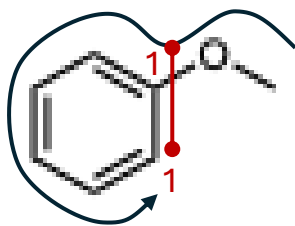
SMILES — Simplified Molecular-Input Line-Entry System



CCC(=O)O



OC(=O)CC



COc1ccccc1



C1=C[CH]C=C1.C1=C[CH]C=C1.[Fe]

SMILES Manipulation



OpenBabel



RDKit

Open-Source Cheminformatics
and Machine Learning

String base representations for small molecules: beyond SMILES



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caffeine

Advanced Search ▾

Search History ↺

How to Search ☺

Treating this as a text search.

BEST COMPOUND MATCH



caffeine; 58-08-2; Guaranine; 1,3,7-Trimethylxanthine; Methyltheobromine; ...

Compound CID: 2519

MF: $C_8H_{10}N_4O_2$ MW: 194.19 g/mol

IUPAC Name: 1,3,7-trimethylpurine-2,6-dione

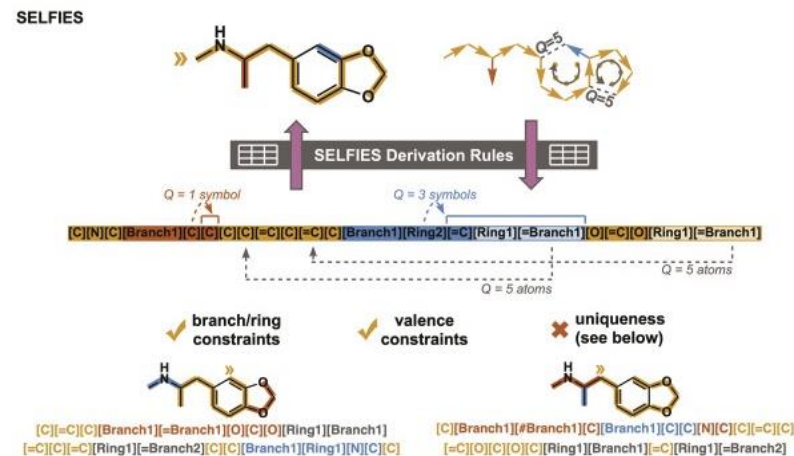
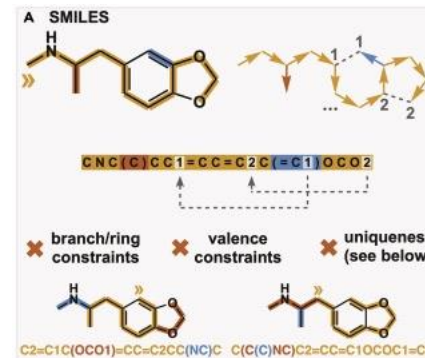
SMILES: CN1C=NC2=C1C(=O)N(C(=O)N2C)C

InChIKey:  RYYVLZVUVIJVGH-UHFFFAOYSA-N

InChI:  InChI=1S/C8H10N4O2/c1-10-4-9-6-5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3

Create Date: 2004-09-16

[Summary](#) [Similar Structures Search](#) [Related Records](#) [PubMed \(MeSH Keyword\)](#)



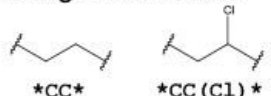
Krenn, M.; Häse, F.; Nigam, A.; Friederich, P.; Aspuru-Guzik, A. *Mach. Learn.: Sci. Technol.* **2020**, 1 (4), 045024.

<https://pubchem.ncbi.nlm.nih.gov/>

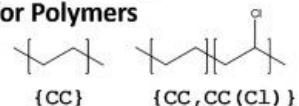
String base representations beyond small molecules

Polymers (BigSMILES)

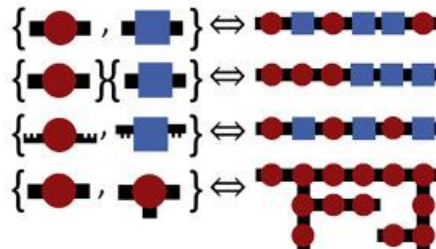
SMILES Representation
for Organic Molecules



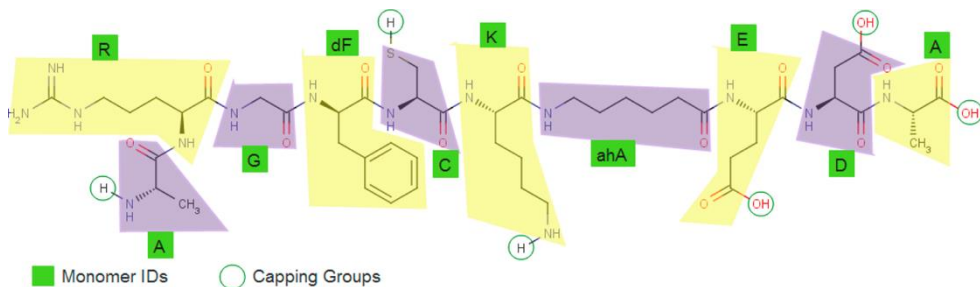
BigSMILES Representation
for Polymers



BigSMILES Supports
a Wide Range of Structures



Proteins (HELM)



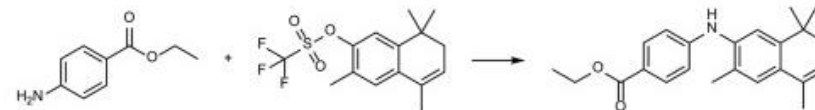
A.R.G.[dF].C.K.[ahA].E.D.A

Materials

SLICES (Simplified Line-Input Crystal-Encoding System)

Wyckoff-position strings

Reactions



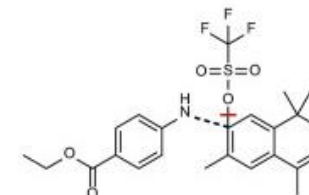
CCOC(=O)c1ccc(N)cc1.CC1=CCC(C)(C)c2cc(OS(=O)(=O)C(F)(F)F)c(C)cc21

>>CCOC(=O)c1ccc(Nc2cc3c(cc2C)C(C)=CCC3(C)C)cc1

Reactants: CCOC(=O)c1ccc(N)cc1.CC1=CCC(C)(C)c2cc(OS(=O)(=O)C(F)(F)F)c(C)cc21

Product: CCOC(=O)c1ccc(Nc2cc3c(cc2C)C(C)=CCC3(C)C)cc1

Condensed graph of reaction (requires atom-mapping):



FC(F)(F)S(=O)(=O)O[->.]c1(cc2C(C)(C)CC=C(c2cc1C)C)[.->]Nc3ccc(cc3C)(OCC)=O

broken bond

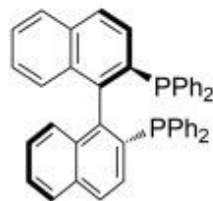
formed bond

Ongoing issues with string based representations

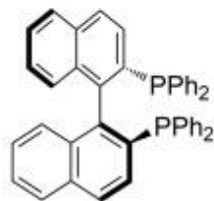
Stereoisomers



[7]helicene



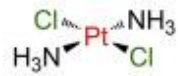
(*R*)-BINAP



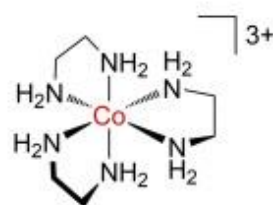
(*S*)-BINAP



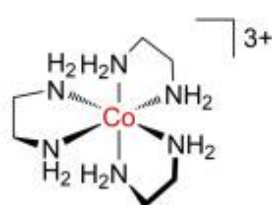
cisplatin



transplatin

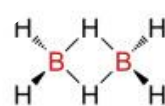


Δ -isomer

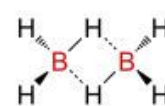


Δ -isomer

Hapticity



1



2



3



4



Fe²⁺

5



6



7

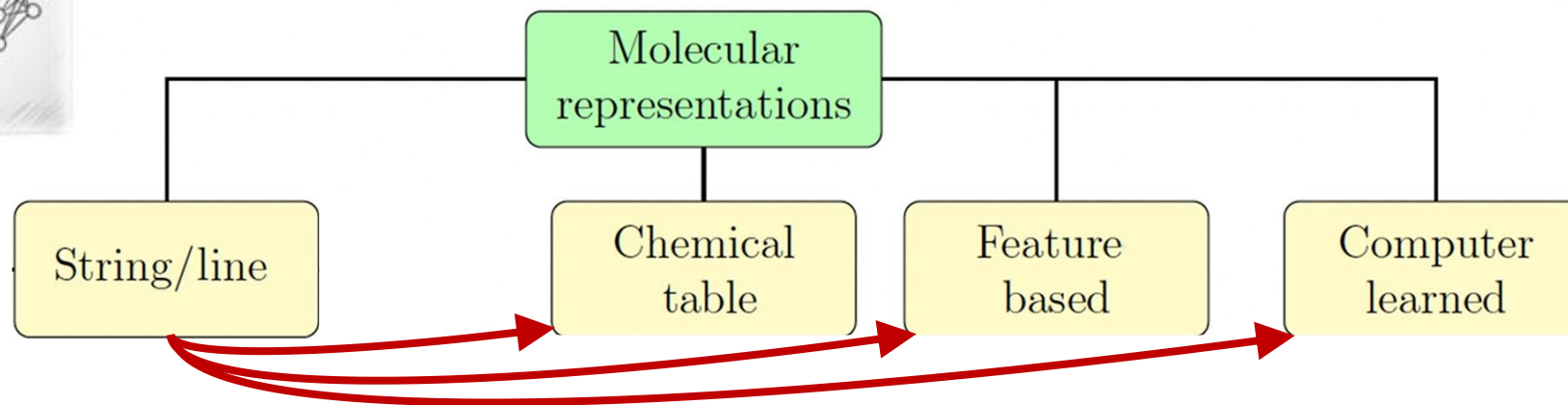
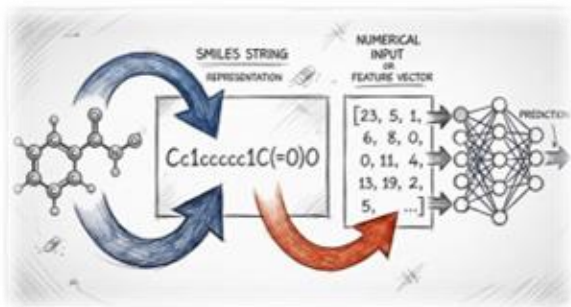


8

Be aware of representation limitations!

Which information is/is not captured in my representation?

From String Based Representation to Numeric Representations



Numerisation/Vectorization of
the Information

In the literature descriptors or features are both used for “numerical representations”

Descriptors Survey

I. Small Molecule Descriptors

- I. Lipinski
- II. Fingerprints
- III. Experimentally Derived Descriptors
- IV. Computed Descriptors
- V. Predicted Descriptors

II. Reaction Descriptors

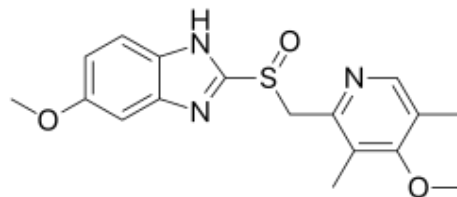
- I. Fingerprints
- II. Energy Surfaces
- III. Concatenation of Molecular Descriptors

Numeric representations: Small Molecules

Lipinski's rule of 5 for drug likeness:

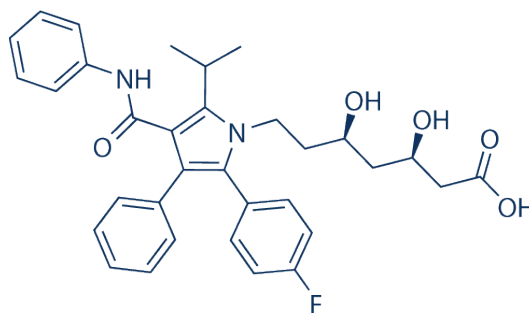
- No more than 5 hydrogen bond donors
- No more than 10 hydrogen bonds acceptors
- Molecular mass less than 500 daltons
- Partition coefficient water-octanol (ClogP) < 5

Omeprazole



- 1
- 5
- 345 g/mol
- 2.3 – 2.6

Atorvastatin



- 4
- 56
- 559 g/mol
- 2.2 – 2.56

Descriptors Survey

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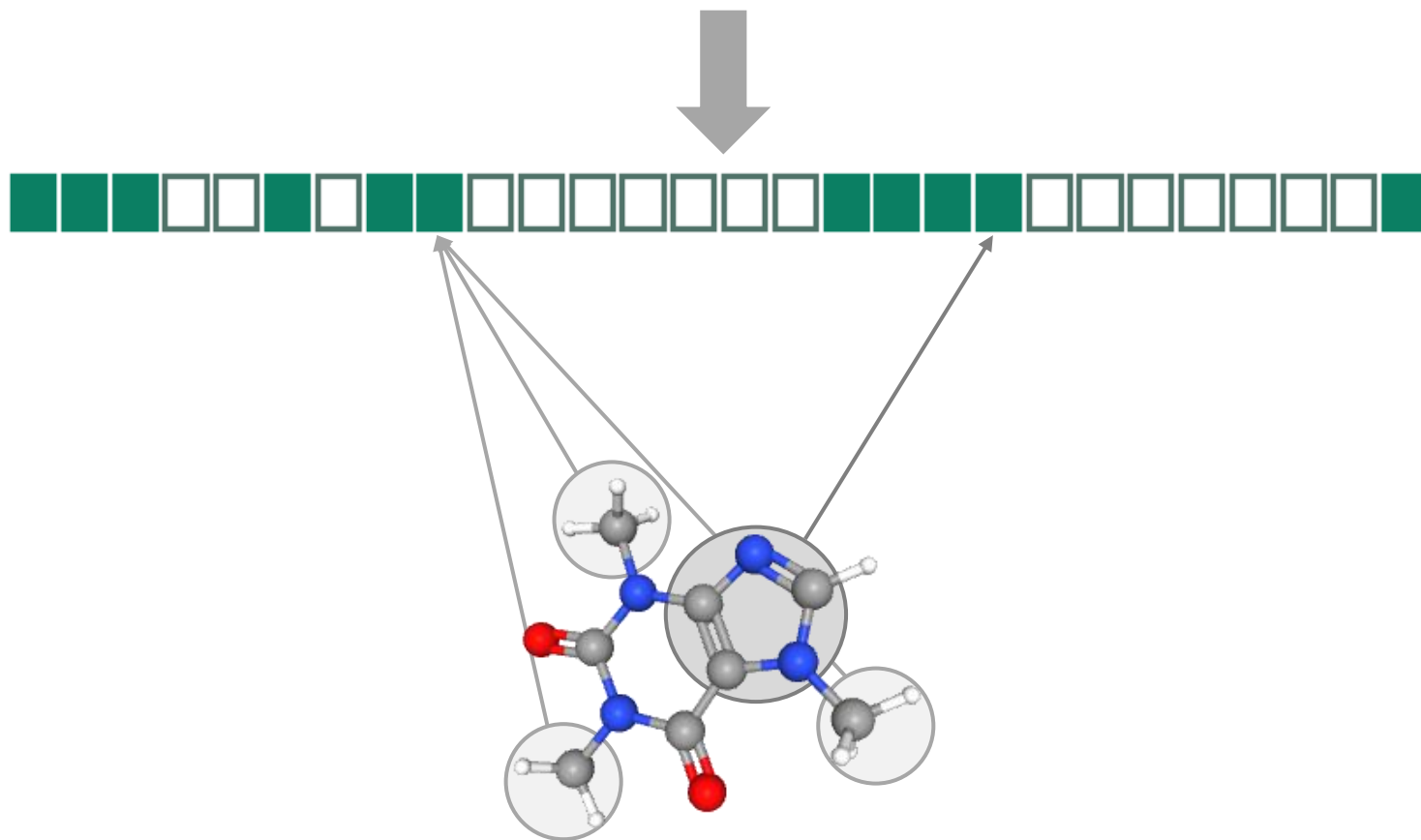
II. Reaction Descriptors

- I. Fingerprints
- II. Energy Surfaces
- III. Concatenation of Molecular Descriptors

Numeric representations: Small Molecules

Molecular Fingerprints Developed for Quantitative Structure Activity Relationships (QSAR):

CN1C=NC2=C1C(=O)N(C(=O)N2C)C



Advantages:

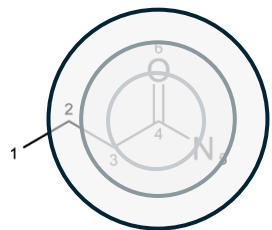
- Very fast to compute
- Tanimoto/Jaccard similarity

Information encoded:

- 2D structure/connectivity
- Stereochemistry
- No 3D information
- No conformers

Numeric representations: Small Molecules

Extended Connectivity Fingerprints (ECFPs):



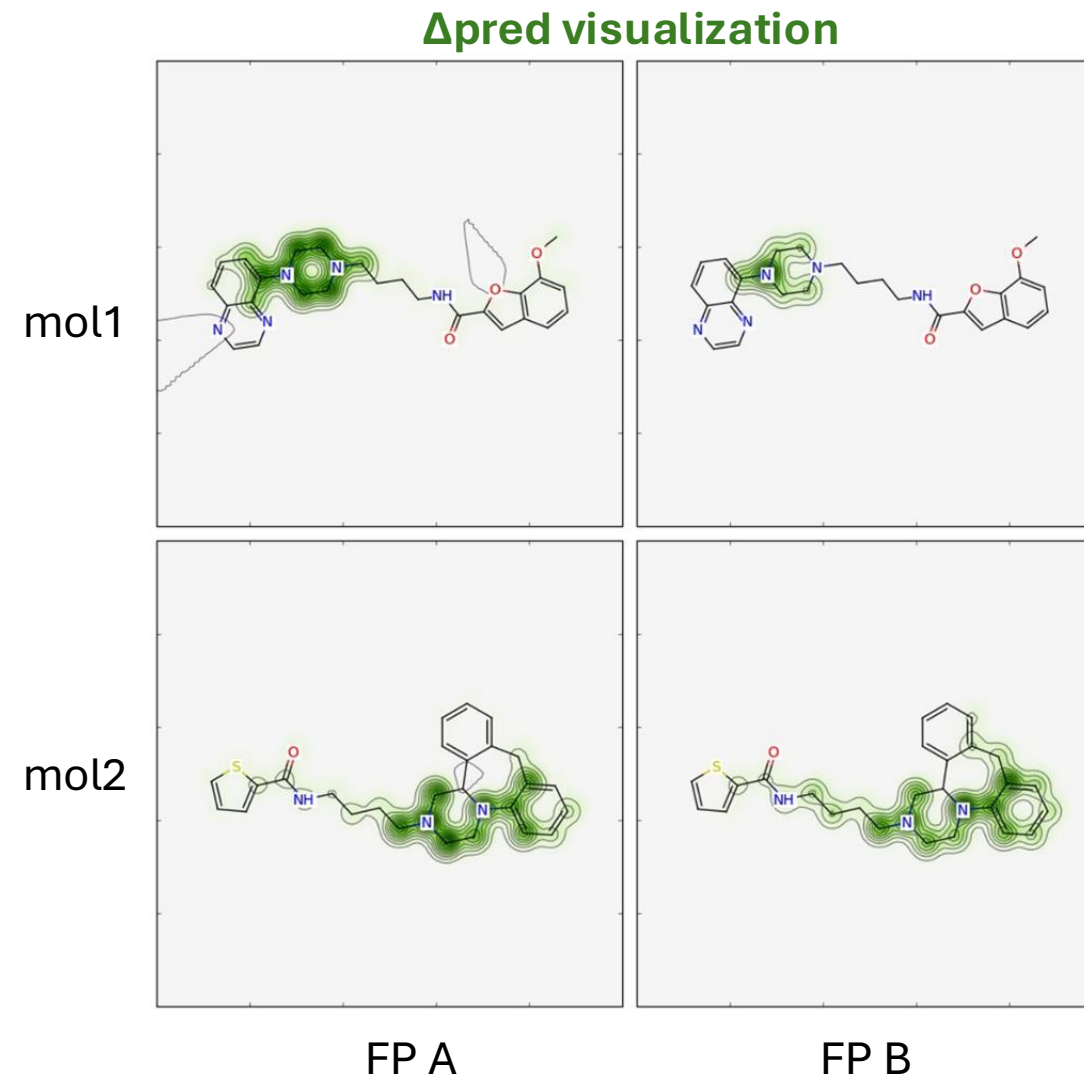
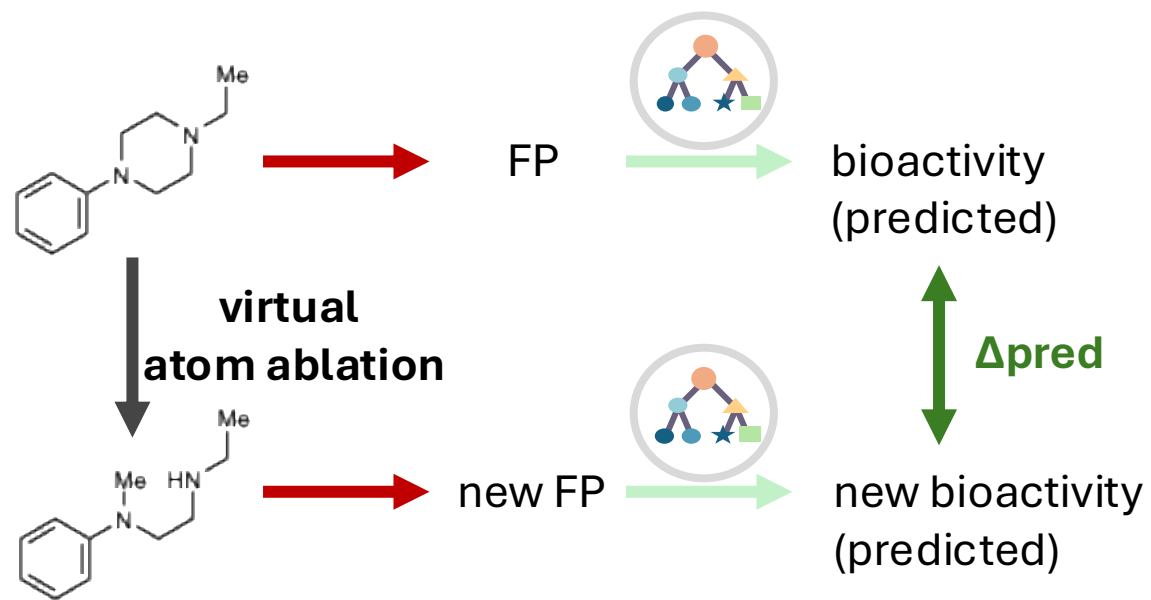
Radius 0

Radius 1



Radius 2

Application ECFPs for QSAR



Recap on Molecular Fingerprints

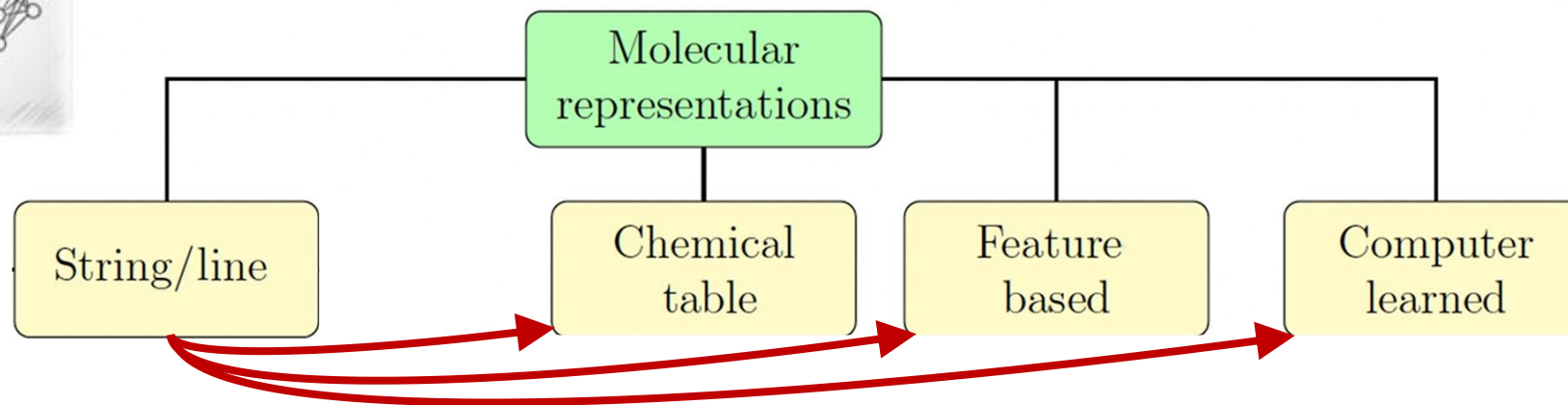
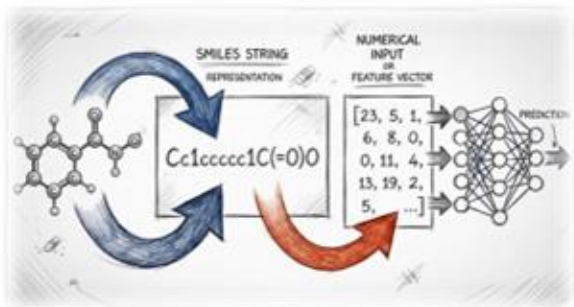
When to use:

- Good for a fast baseline small molecules activity
- Efficient with relatively small datasets (100 – 10k molecules)

When not to use:

- Properties/Activities depend on 3-D shape or conformers
- Very small datasets
- Non-organic molecules/other chemical systems

From String Based Representation to Numeric Representations



Numerisation/Vectorization of
the Information

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Feature Based Representations: Experimental Descriptors

Hammett descriptors:

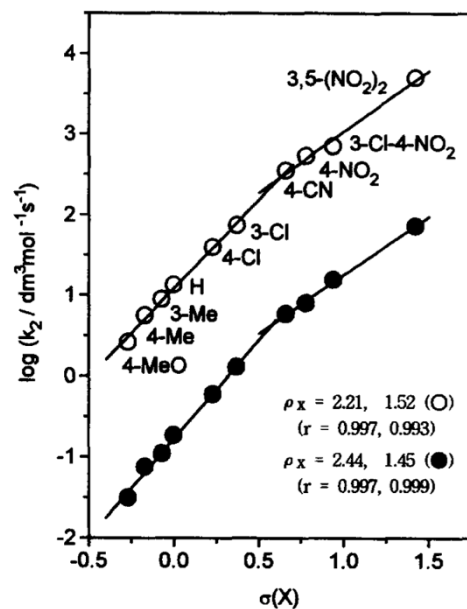


Figure 1. Hammett plots for the reactions of *p*-nitrophenyl *X*-substituted benzoates with OH⁻ (○) and *p*-chlorophenoxide (●) in H₂O containing 20 mole % DMSO at 25.0 ± 0.1 °C.

Advantages:

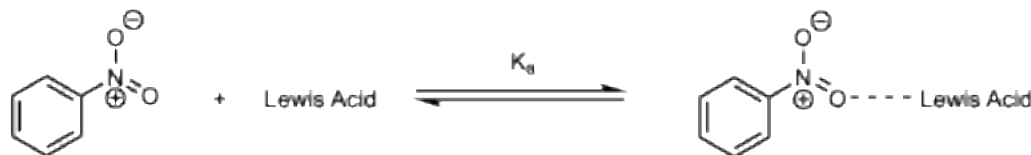
- Mechanism interpretability
- Condensed information
- Small dataset interpretability

Limitations:

- Specific electronic effects
- Availability

Feature Based Representations: Experimental Descriptors

Predicting Lewis Acid Base Affinity with Experimental Descriptors



NMR Acceptor Number (AN)
Water Exchange Rate (WERC)
 pK_h
Electrostatic Acceptor Number (E_a)
Covalent Acceptor Number (C_a)
Infrared Xanthone Complexes (IrX)
Infrared TPPO Complexes (IrTPPO)
Infrared EthylAcetate Complexes (IrAcOEt)
Oxidation Peak (Ox)
Complexing Power (CP)

Advantages:

- Mechanism interpretability
- Condensed information
- Small dataset interpretability

Limitations:

- Specific electronic effects
- Availability of experimental descriptors

Today:

- Computed descriptors

Descriptors Survey

I. Small Molecule Descriptors

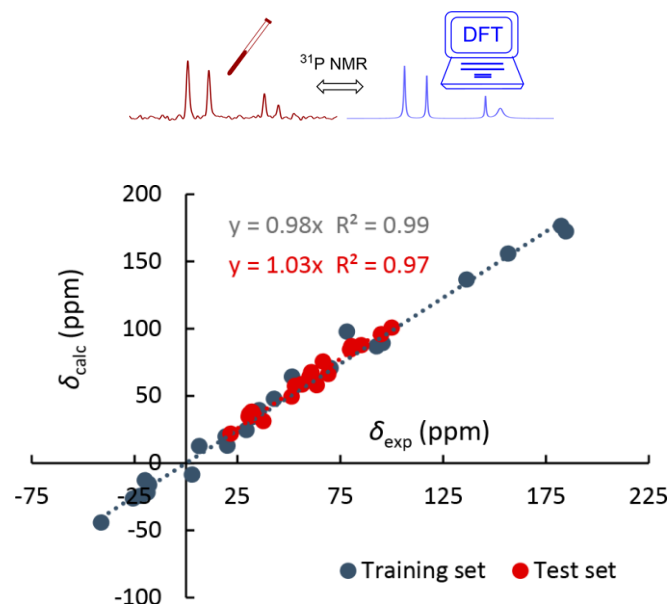
- I. Lipinski
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- V. Predicted Descriptors

II. Reaction Descriptors

- I. Fingerprints
- II. Energy Surfaces
- III. Concatenation of Molecular Descriptors

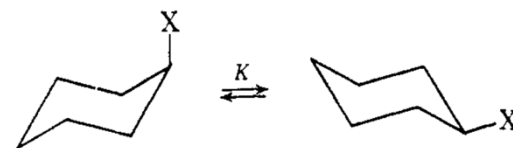
Feature Based Representations: Computed Descriptors

Phosphine Featurization by DFT:



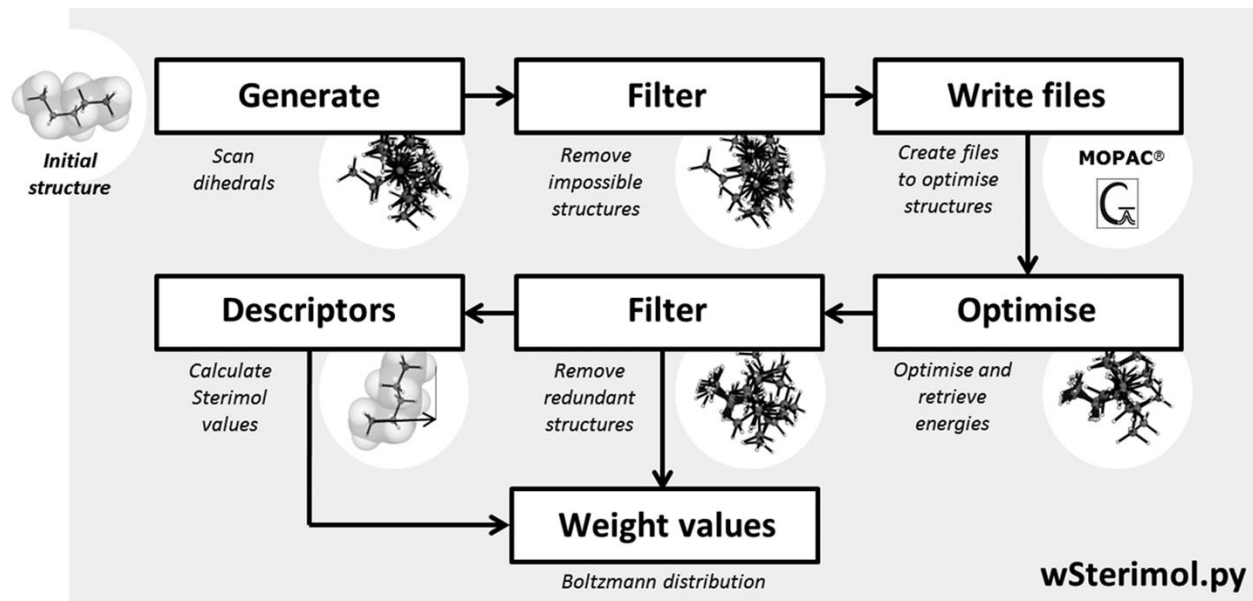
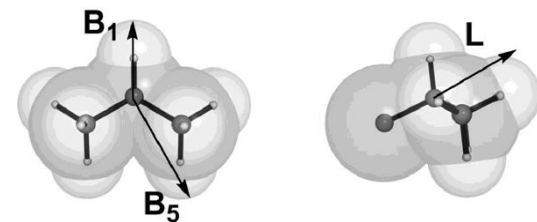
Geometries optimized at the B3LYP/BS1 level of theory. ^{31}P NMR shielding constants were evaluated with the GIAO method at the B3LYP/BS2 level of theory.

Steric Effects:



$$A = -\Delta F^\circ = RT \ln K$$

Sterimol Descriptors:

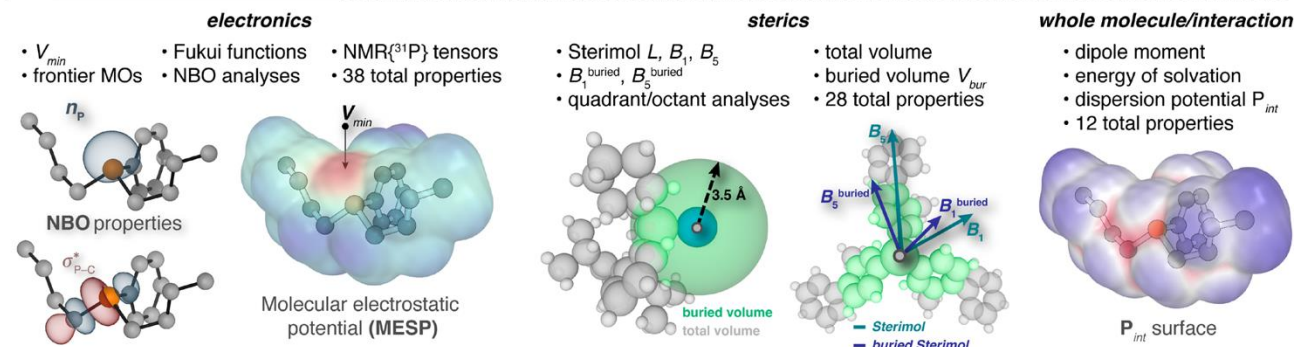


Payard, P.-A.; Perego, L. A.; Grimaud, L.; Ciofini, I. *Organometallics* **2020**, 39 (17), 3121–3130.
Jensen, F. R.; Bushweller, C. H.; Beck, B. H. *J. Am. Chem. Soc.* **1969**, 91(2), 344–351
Brethomé, A. V.; Fletcher, S. P.; Paton, R. S. *ACS Catal.* **2019**, 9 (3), 2313–2323.

Feature Based Representations: Descriptors Libraries

1 evaluating conformer properties

78 properties are evaluated for **each conformer** in the representative ensembles



2 ensembling properties into ligand descriptors

Number of ligand descriptors does not depend on the number of conformers

conformer properties → ligand descriptors

Boltzmann-weighted conformer average

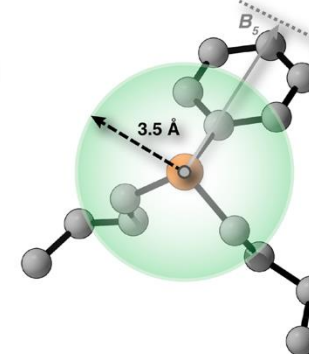
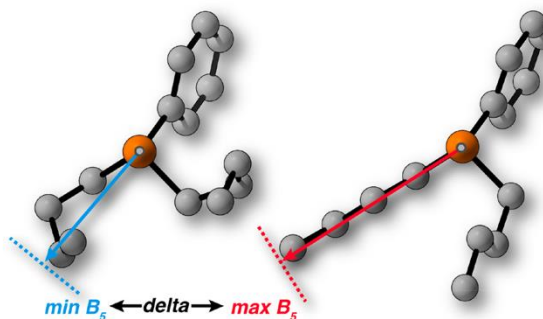
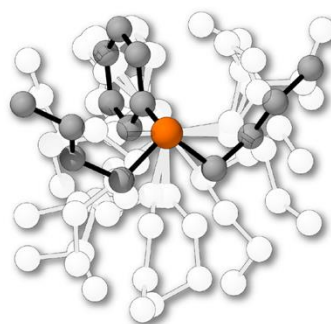
- a representative value for entire ensemble

min-max ligand size span

- denotes the fundamental steric extrema
- the ligand can assume any accessible property value within this range

conformer with smallest V_{Bur}

- a representative conformer for sterically congested complexes



Conformer ensemble generation

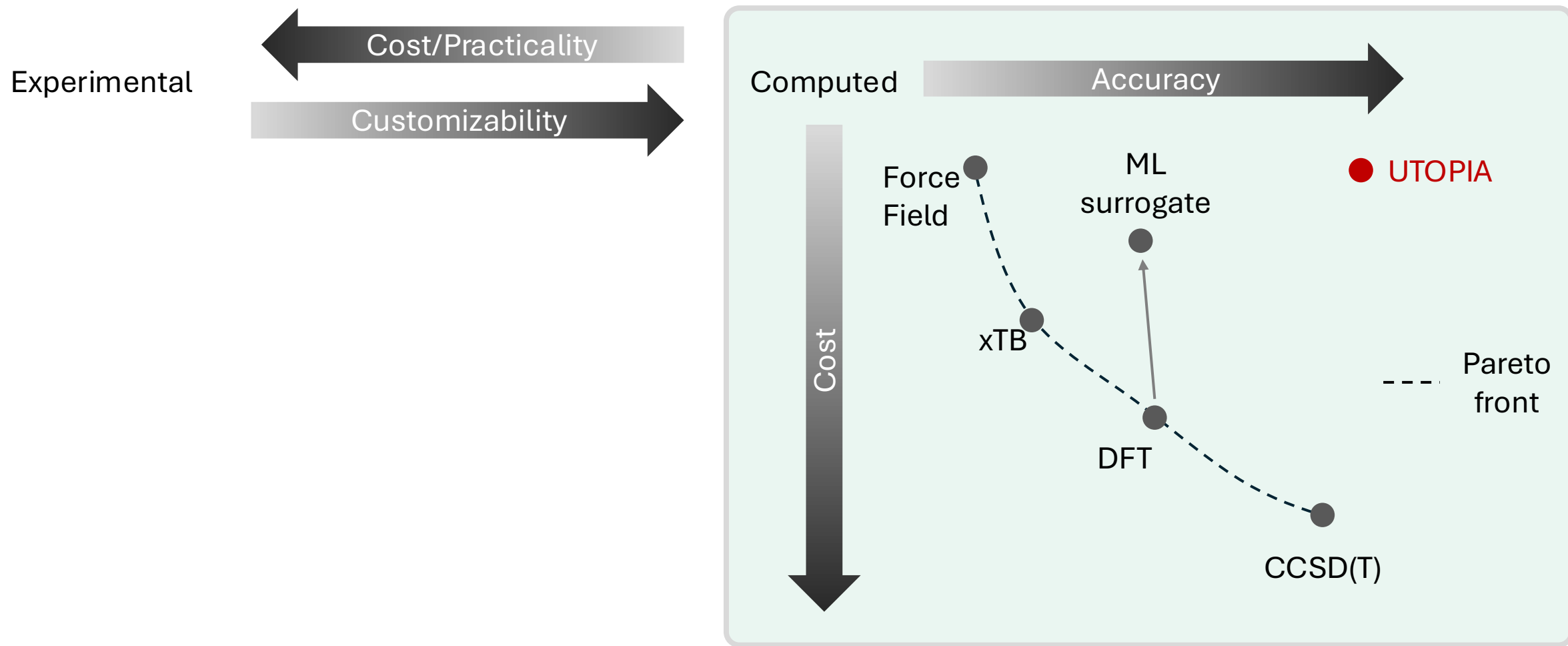
organophosphorus compounds



1,558 ligands

Descriptor Database

Feature Based Representations: Towards Predicted Descriptors



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- III. Concatenation of Molecular Descriptors

Examples of ML-Surrogates For Descriptor Generation

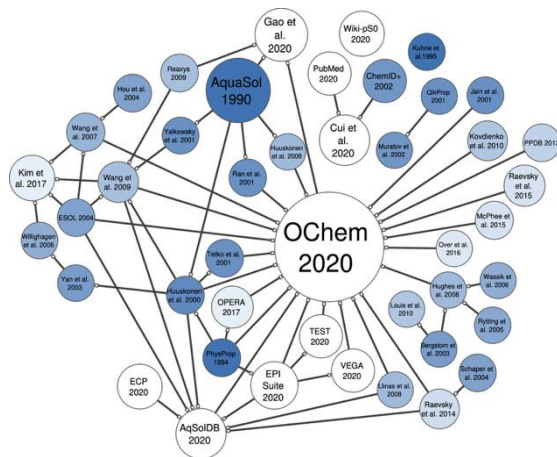
MLIPs for Geometry Optimization (Wednesday!)

- UMA
- MACE
- AIMNET

BDE predictions:

- BondNET charged species
- Alfabet (homolytic neutral molecules)

Solubility:



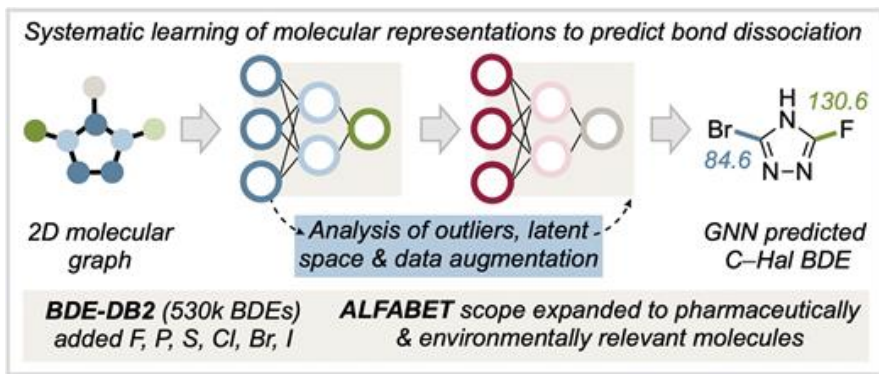
pKa predictions:

MolGpKa (<https://xundrug.cn/molgpka>)

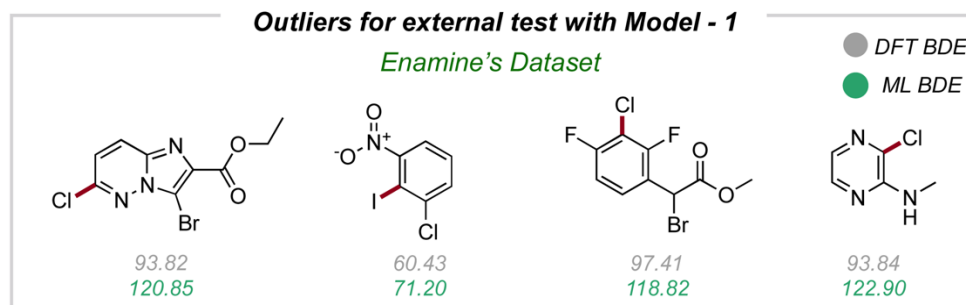
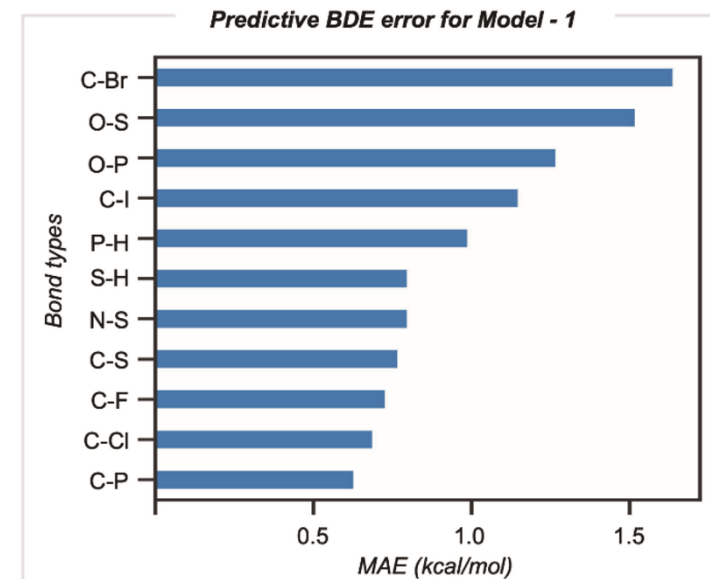
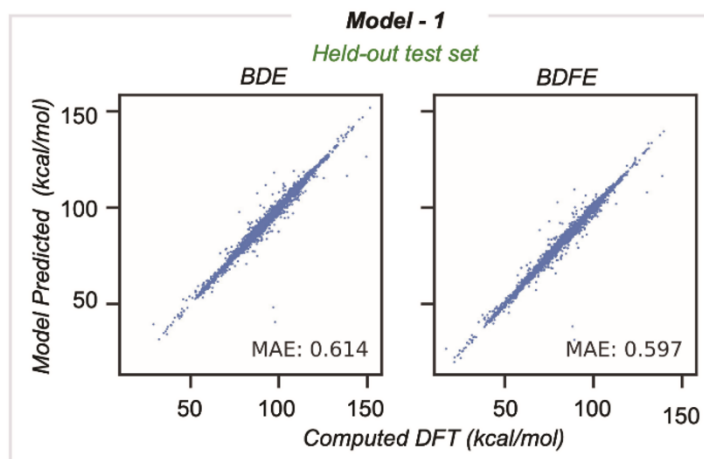
Expert Crafted Descriptors: Using ML models

Prediction of BDEs using ML:

ML pipeline:



Validation and Domain of Applicability

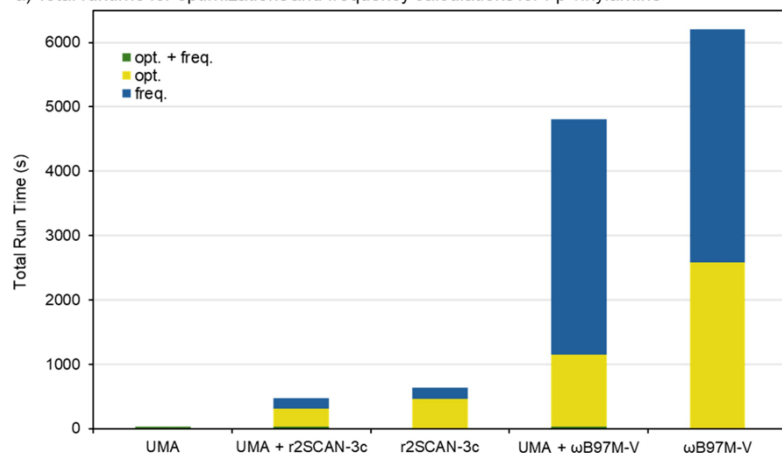


Always evaluate validity
Recompute some data points with the high accuracy method for important points!

Expert Crafted Descriptors: Using ML models

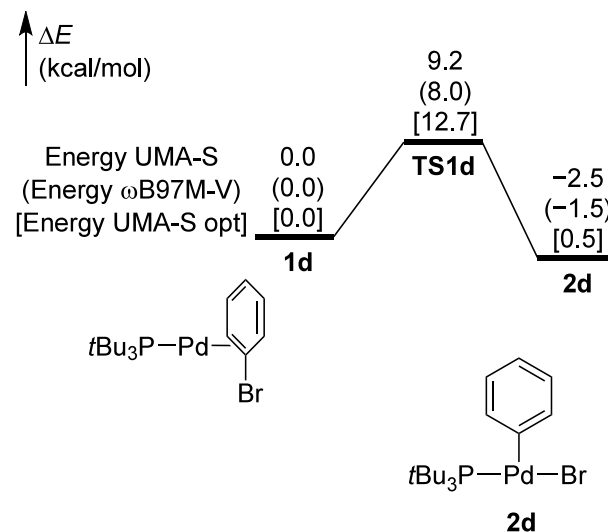
Machine Learning Interatomic Potentials as DFT surrogate (Structure Optimization/Energy Calculation):

a) Total runtime for optimizations and frequency calculations for Fp-vinylamine

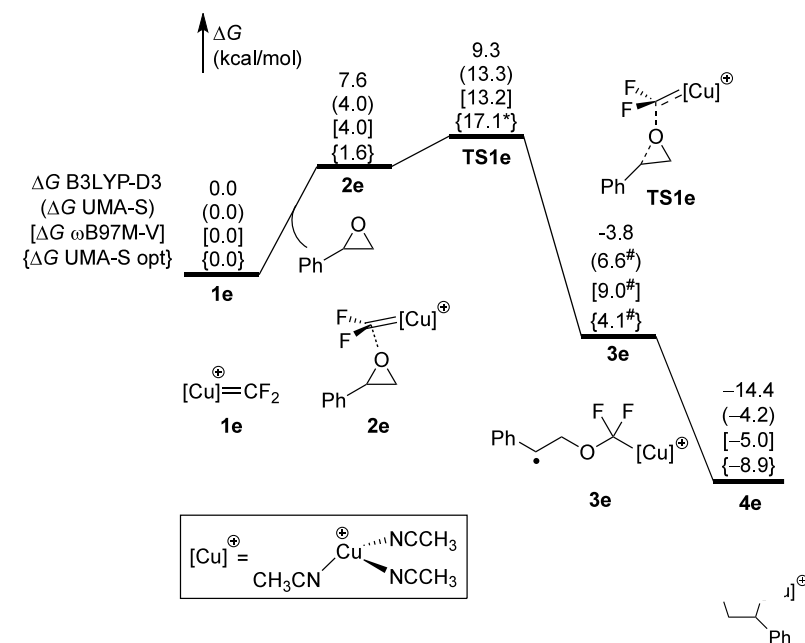


>99% reduction compute

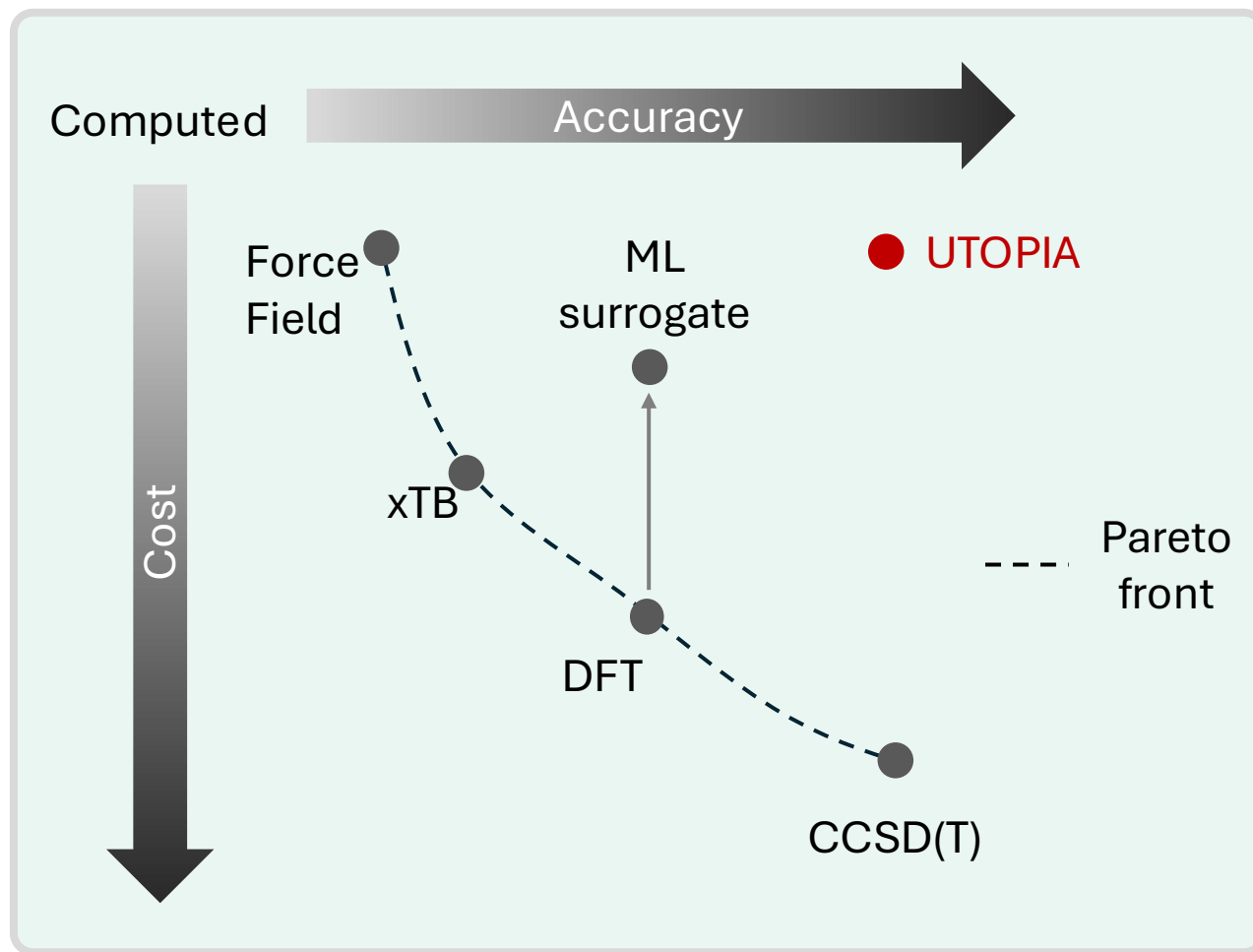
Closed shell



Open shell



Expert Crafted Descriptors: Using ML models



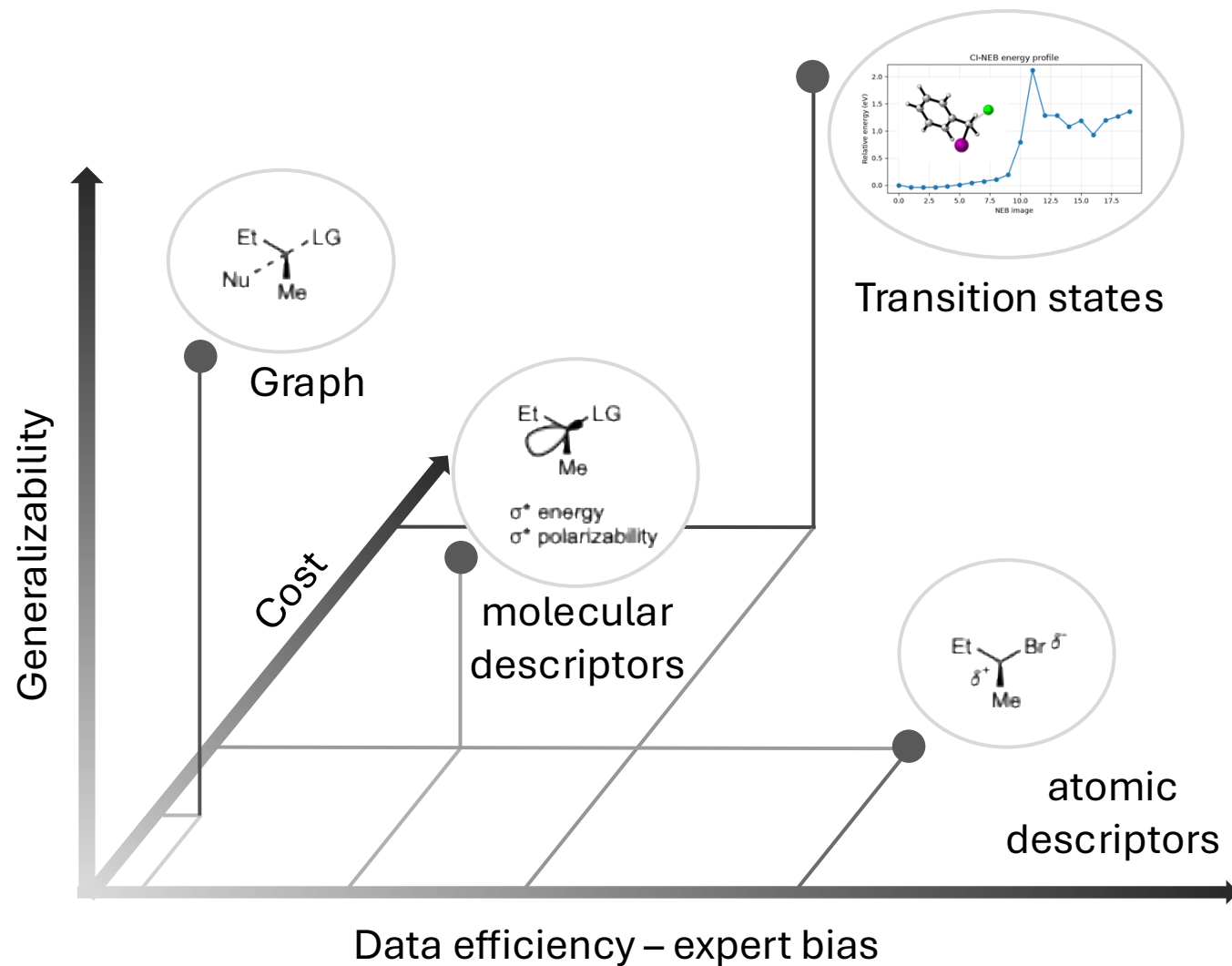
Ab-initio or semi-empirical computational tools have identifiable flaws/literature experience.

ML surrogates are less predictable: Good usage practice:

- Benchmark few points vs DFT/reference method
- Understand/evaluate domain of applicability

Expert-Crafter Descriptors: Precision/Generalizability Trade-off

Descriptors for modeling rates of S_N2 ?

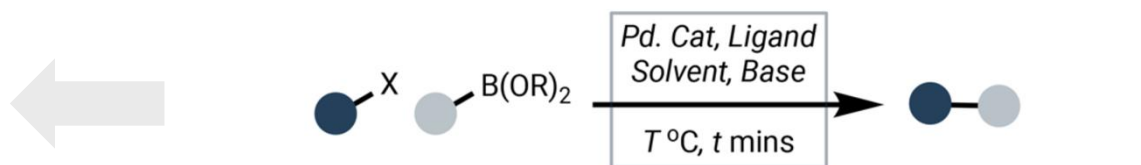


Multi-parameter/Combinatorial Chemical Systems

Example of Organic Reaction Conditions

a. What Are Reaction Conditions?

Concatenate
descriptors of
reactant, conditions
and product



The components 'above the arrow'
which facilitate a chemical reaction

Encode the reaction
as a whole

Multi-parameter/Combinatorial Chemical Systems: Organic Reactions

Encode the reaction as a whole:

Reaction SMILES:



Differential Reaction Fingerprints

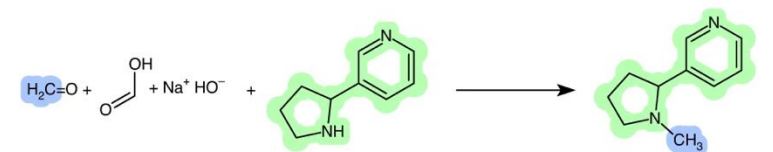
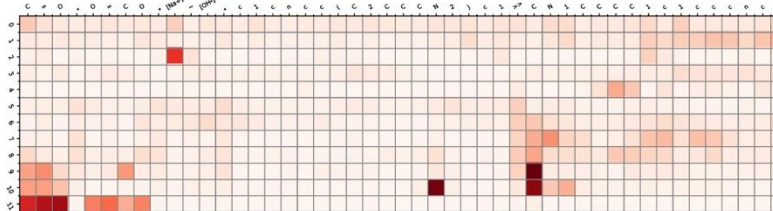


Learned Encoding (Transformers)

Eschweiler–Clarke methylation [1.2.4]

C=O.O=CO.[Na+].[OH-].c1cncc(C2CCCN2)c1>>CN1CCCC1c1ccccc1

BERT: [CLS] attention per layer



Seq2Seq: encoder-decoder attention

Advantages:

- Universal
- Interpretable (DRFP)

Disadvantages:

- Coarse grained information
- Requires large datasets

Schneider, N.; Lowe, D. M.; Sayle, R. A.; Landrum, G. A. *J. Chem. Inf. Model.* **2015**, 55 (1), 39–53. / Probst, D.; Schwaller, P.; Reymond, J.-L. *Digital Discovery* **2022**, 1 (2), 91–97.

Schwaller, P.; Laino, T.; Gaudin, T.; Bolgar, P.; Hunter, C. A.; Bekas, C.; Lee, A. A. *ACS Cent. Sci.* **2019**, 5 (9), 1572–1583. /

Multi-parameter/Combinatorial Chemical Systems: Organic Reactions

Encode the reaction as a whole:

Reaction SMILES:



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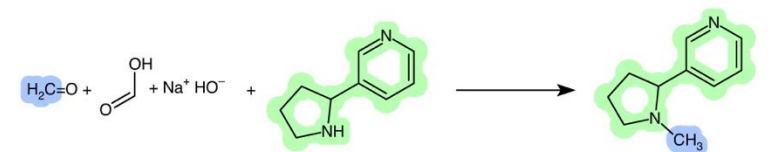
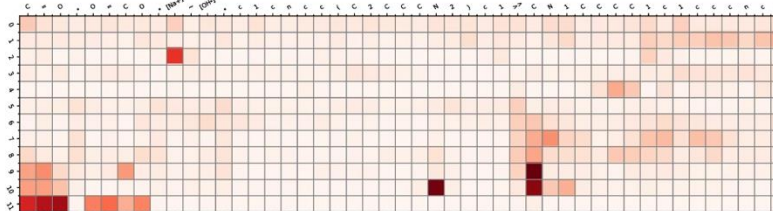


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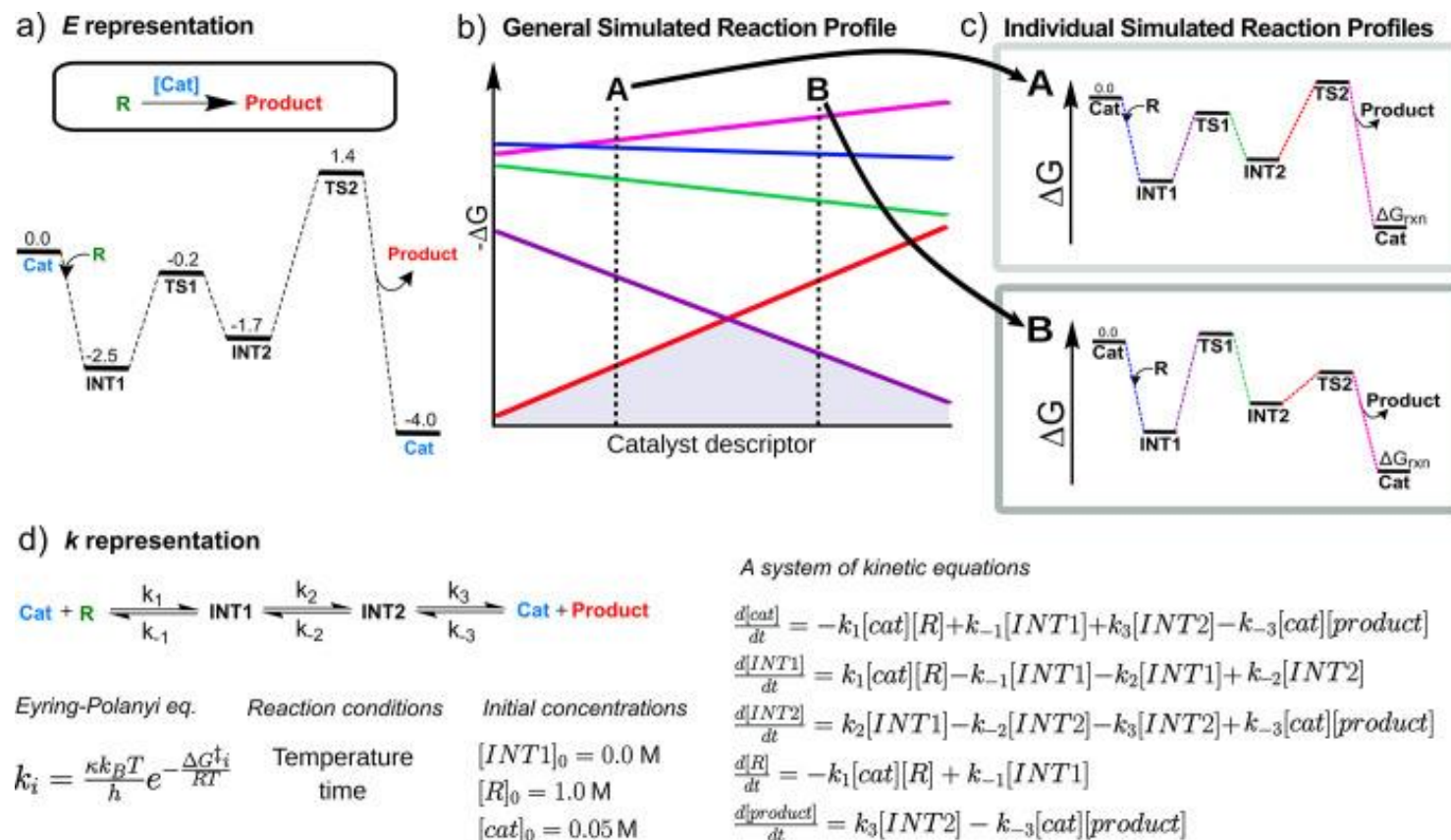
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Multi-parameter/Combinatorial Chemical Systems: Organic Reactions

Encode the reaction as a whole:



Advantages:

- Captures subtle reactivity trends
- Transferable between catalysts

Disadvantages:

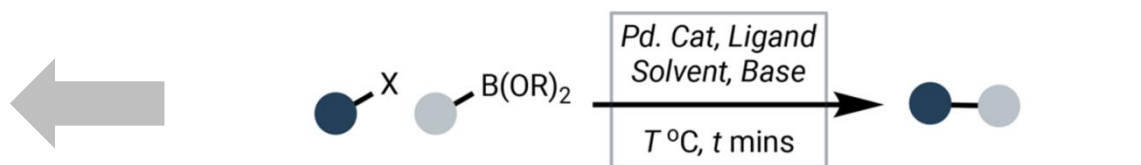
- Mechanism must be known/constant
- Computationally costly
- Only encodes catalyst

Multi-parameter/Combinatorial Chemical Systems

Example of Organic Reaction Conditions

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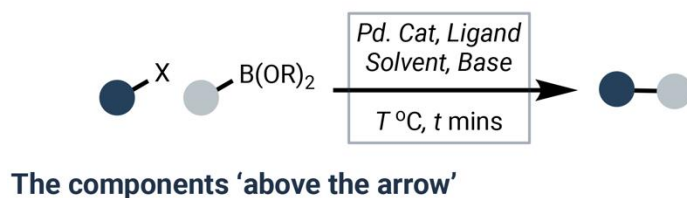


The components 'above the arrow'
which facilitate a chemical reaction

Encode the reaction
as a whole

Multi-parameter/Combinatorial Chemical Systems: Organic Reactions

Integrating multiple variables?



c. Mathematical Formulation

REAGENTS

CHEMICAL VARIABLES

Reacting species which **don't contribute a heavy atom** to the product.

- *Pd Cat.*
- *Ligand*
- *Solvent*
- *Base*

Categorical

PHYSICAL PARAMETERS

NON-CHEMICAL VARIABLES

Other non-chemical variables that influence a reaction.

Here:

- *Temp.*
- *Time*
- + more...

Continuous

Concatenation of features:

	solvent	base	additive
	■ ■ ■ □ □ ■	□ □ ■ □ ■ ■ □ □	□ □ □ □ ■

Choice of features: choosing the right amount of expert bias to use?

Bias into Engineered descriptors?

Example of solvent encoding

One-Hot Encoded Vectors

Using 'exact' reagent identities



MeOH

One-Hot Encoded Vectors

Using reagent 'classes'



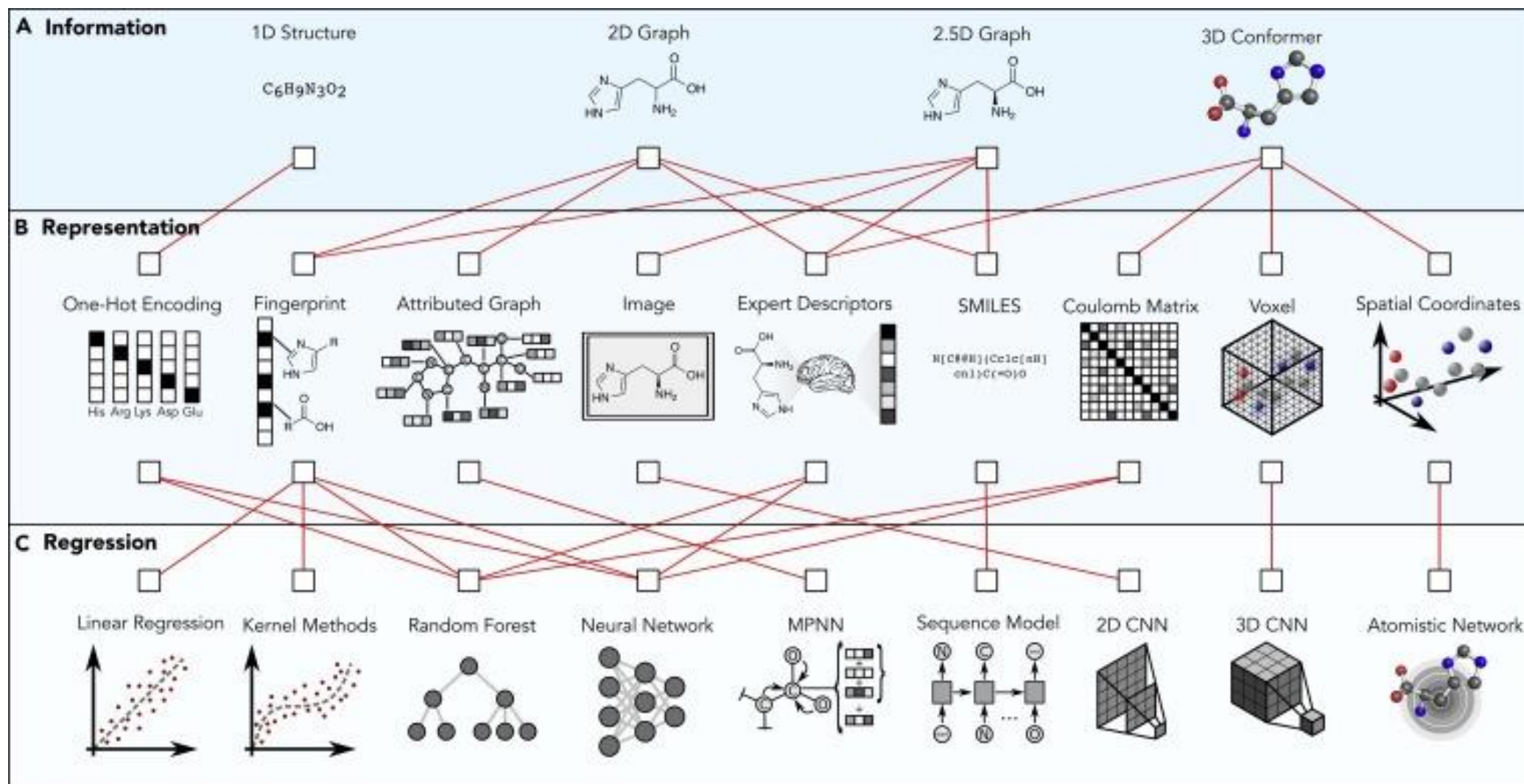
Continuous Representation

Physical Properties,
Learnt Embeddings etc.



Expert bias / data efficiency (if feature related to property)

Molecular Representations: Outlook



Pattanaik, L.; Coley, C. W. *Chem* **2020**, 6 (6), 1204–1207.

Molecular Representations: Outlook

Three Eras of Molecular Representation

Same regression target — three answers to “where do the features come from?”

1 • Engineered descriptors

Pick features by hand from domain knowledge.

*MW · logP · Sterimol · %Vbur · MACCS /
Morgan fingerprints*

+ interpretable; strong in low-data regimes

– needs insight; may miss the key axis

2 • Supervised end-to-end

Learn the embedding from the graph for one task.

*Message-passing GNNs — ALFABET,
Chemprop*

+ no manual featurization; SOTA with data

– data-hungry; one model per property

3 • Self-supervised pretraining

Pretrain on millions of unlabelled molecules, then fine-tune.

*Foundation models — Uni-Mol (3D),
MolFormer, MoleCLIP*

+ data-efficient transfer; robust to shift

– large, opaque; pretraining cost

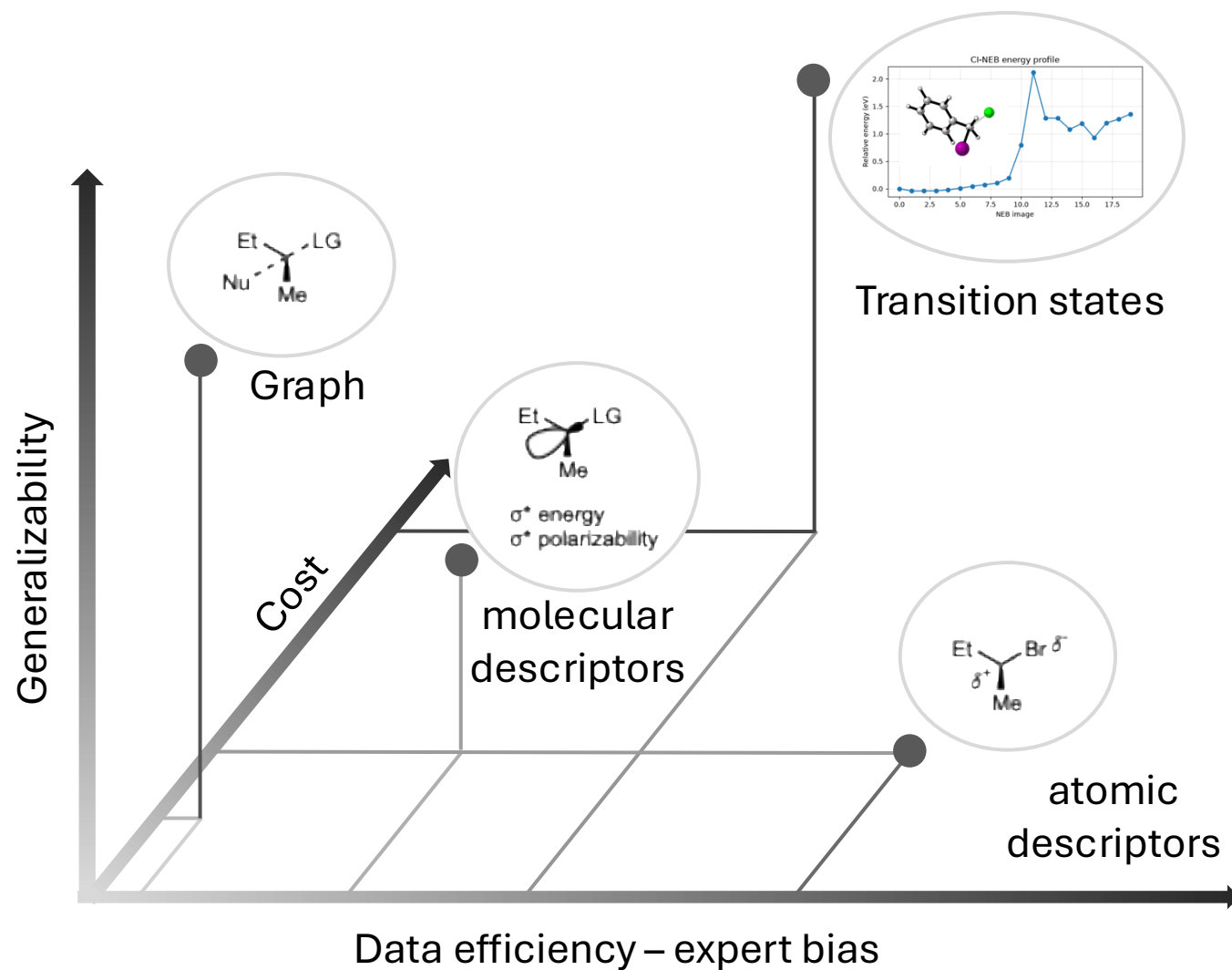
hand-crafted → learned per task → learned once, transferred everywhere

https://github.com/camlcworkshop/camlcworkshop.github.io/blob/master/Sessions/Paton/Paton_2D_3D_Featurization.pdf

Plan

- Evolution of “data-driven” chemistry
- Overview of existing representations
- **Importance of the choice of the representation**
- Data preprocessing: practical considerations
- Data visualization

Expert-Crafter Descriptors: Precision/Generalizability Trade-off



Plan

- Evolution of “data-driven” chemistry
- Overview of existing representations
- Importance of the choice of the representation
- **Data preprocessing: practical considerations**
- Data visualization

Multiple feature preprocessing

- **Standardization**
- **Removing zero-variance features**
- **Removing correlated features**
- **Data visualization: understanding local and global data structure/potentially removing outliers**

Plan

- Evolution of “data-driven” chemistry
- Overview of existing representations
- Importance of the choice of the representation
- Data preprocessing: practical considerations
- **Data visualization**

Principal Component Analysis (PCA)

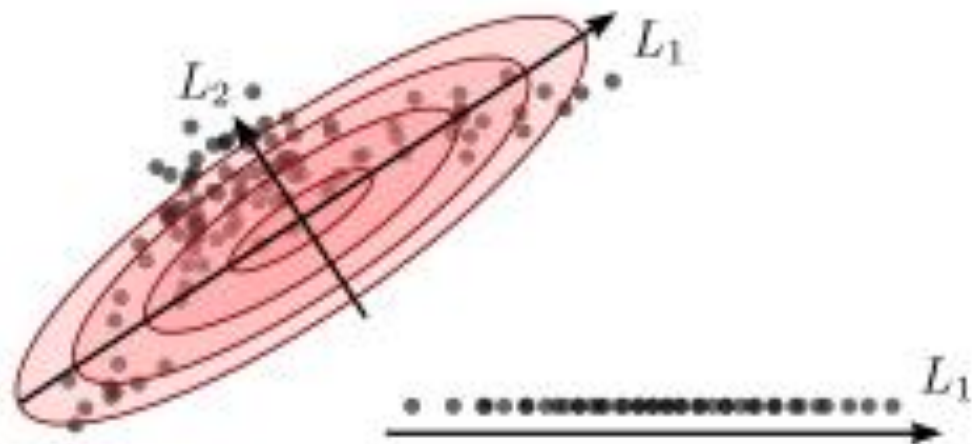
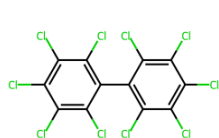


Figure from: <https://github.com/BingqingCheng/ML-in-chemistry-101/blob/master/slides/l5.pdf>

PCA: an example

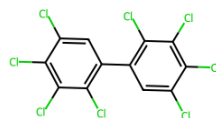
Example molecules



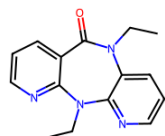
logS = -4.08



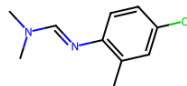
logS = -3.00



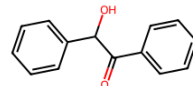
logS = -2.92



logS = 0.09



logS = 0.09



logS = 0.10



logS = 2.10



logS = 2.20

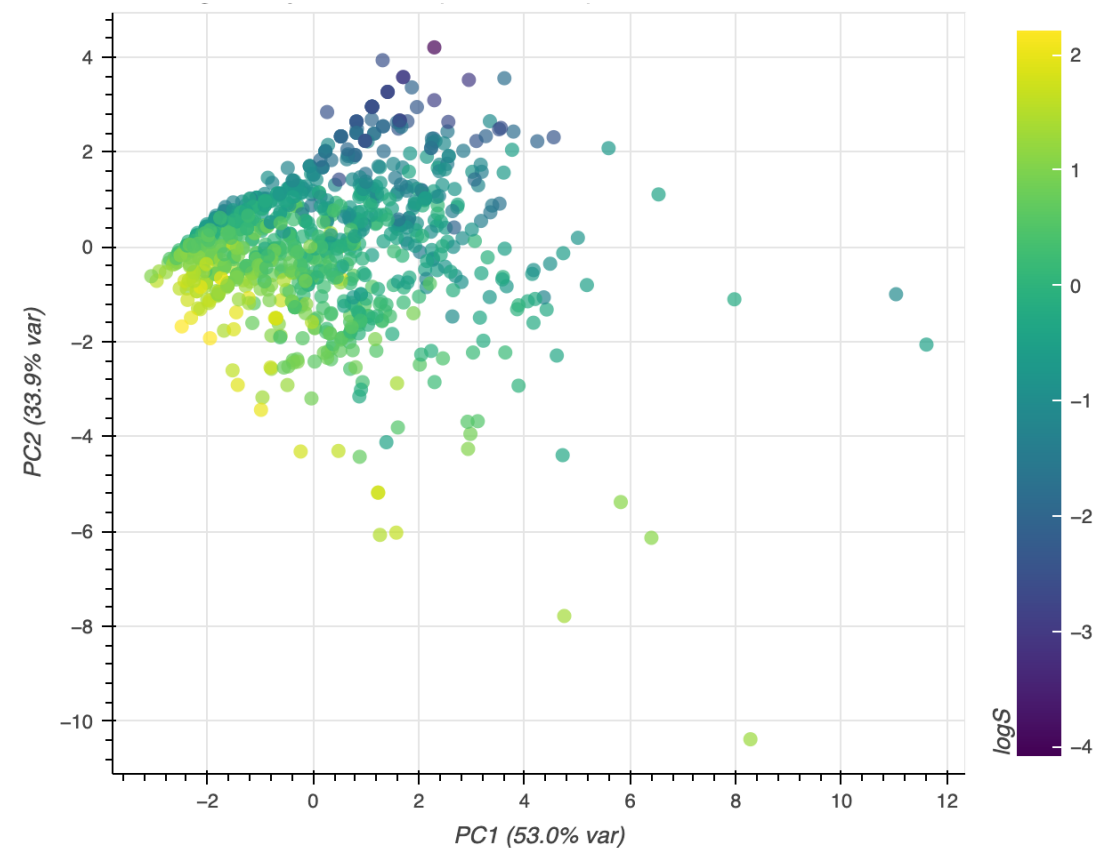


logS = 2.21

Molecular descriptors used:

MolWt, TPSA, #HeavyAtoms, #Ring, MolLopP, #HDonors

PCA projection (PC1 and PC2)



Total variance explained = 86.9%

PCA limitations

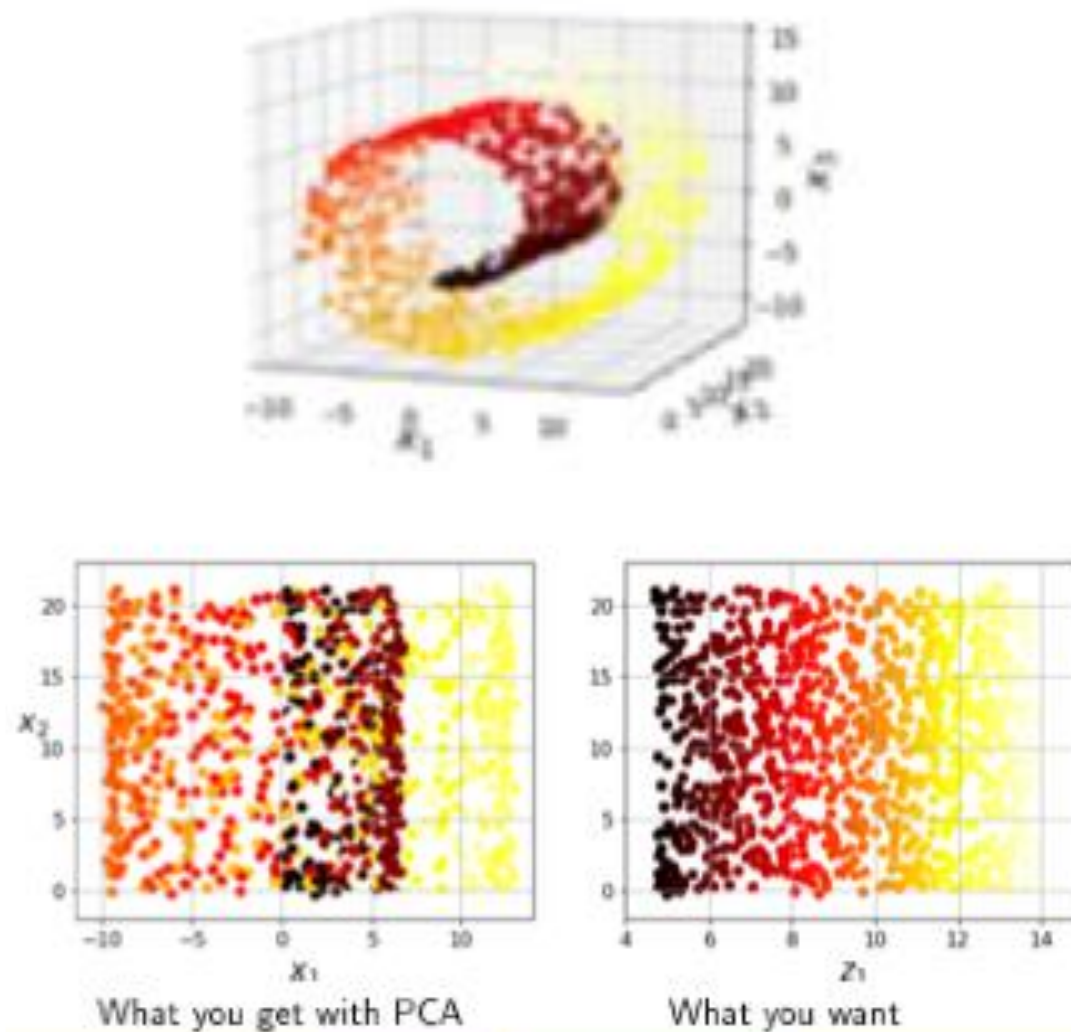
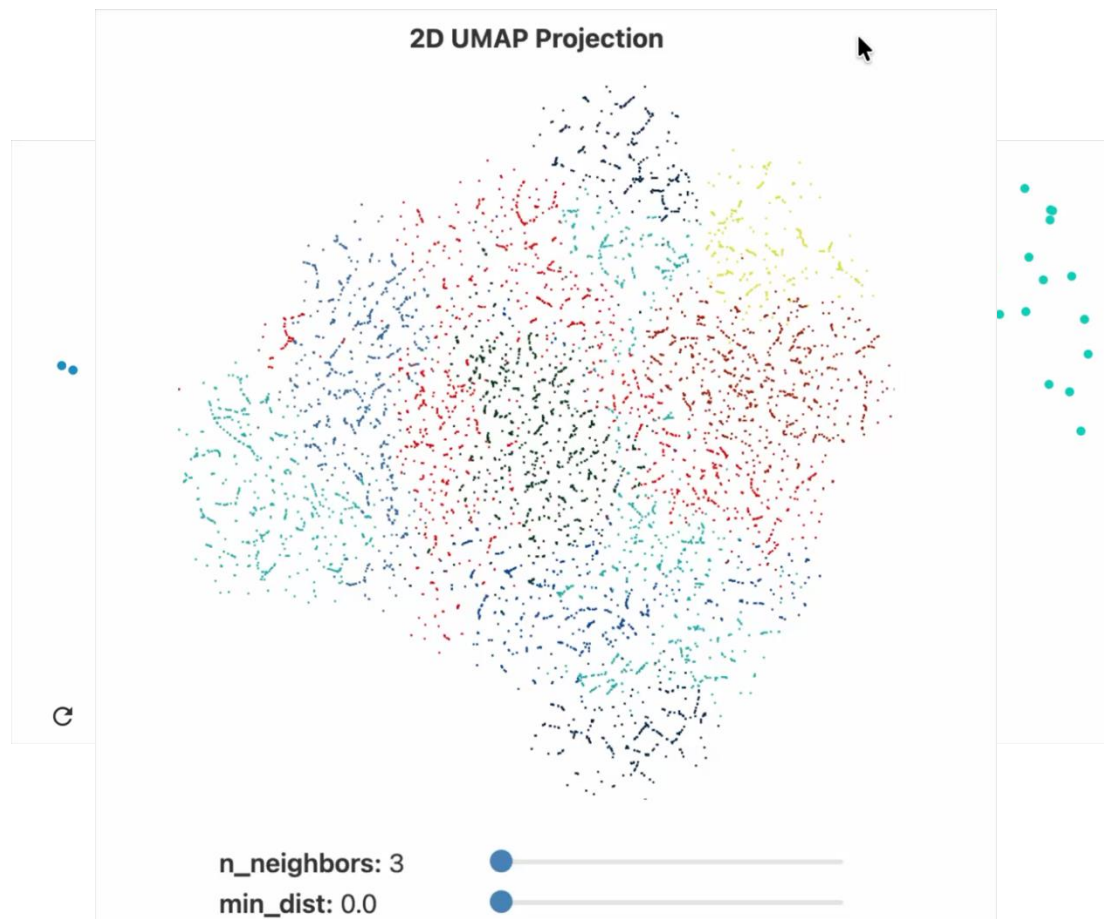


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Beyond PCA for Dimensionality Reduction



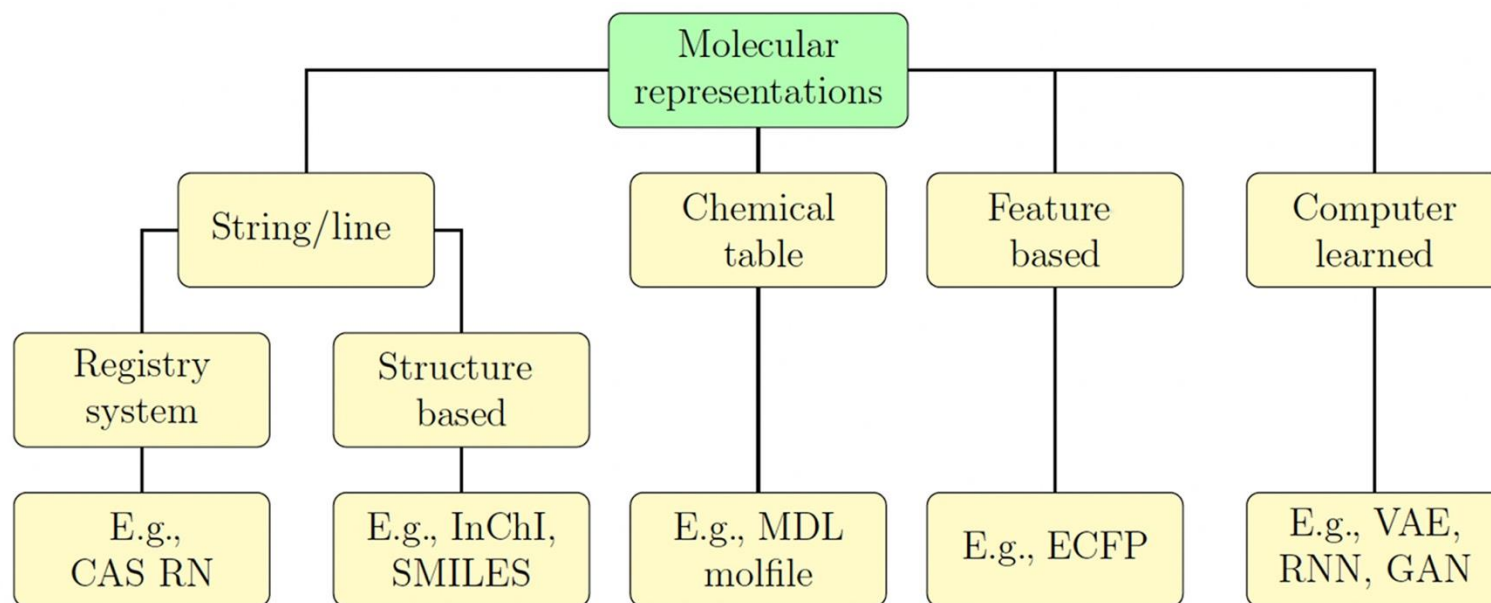
How to preserve local and global structure while reducing dimension?



1. Hyperparameters really matter.
2. Cluster sizes in a UMAP plot mean nothing.
3. Distances between clusters might not mean anything.
4. Random noise doesn't always look random.
5. You may need more than one plot

Conclusion: Do not overinterpret!!!

Future Lectures



- Graphs Neural Networks
- Variational Auto-Encoders
- Transformers

Questions?

